GENERAL ORGANIC CHEMISTRY

Introduction:

This study text develops the relationships between molecular structure, electron distribution, stability and organic reactivity. It covers carbon bonding, electronic and steric effects, reactive intermediates, attacking reagents and the reasoning used in acidity, basicity and mechanism questions.
Read the explanations with the structures beside you. For every comparison, identify the exact atoms, charge, solvent and reaction conditions before applying a memorised order. Worked checks at the end connect the individual effects to examination problems.

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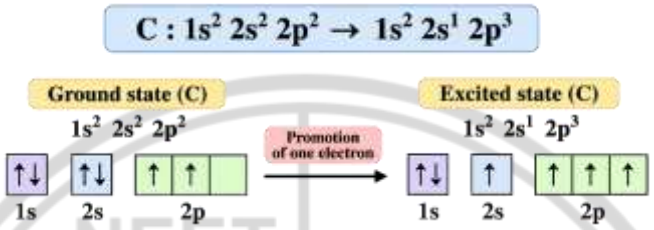
1 Carbon bonding and hybridisation

1.1 Carbon and Its Tetravalency

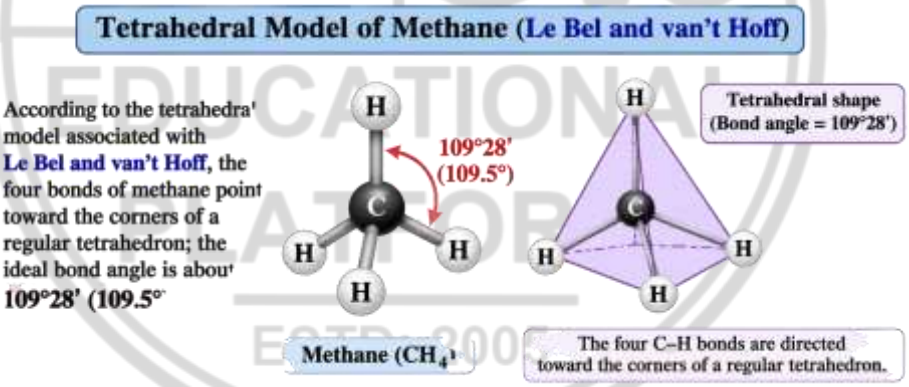
- Carbon (atomic number 6) has four valence electrons. Its ground-state electronic configuration is $1s^2\ 2s^2\ 2p^2$, with two unpaired electrons.

- In the simple valence-bond description, promotion of one 2s electron to an available 2p orbital gives four singly occupied orbitals. Hybridisation then accounts for equivalent bonding directions in methane.

$$C: 1s^2\ 2s^2\ 2p^2 \ \rightarrow \ 1s^2\ 2s^1\ 2p^3$$



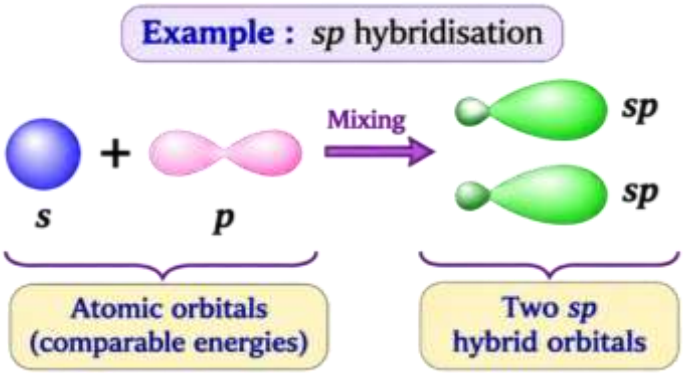
- According to the tetrahedral model associated with **Le Bel and van't Hoff,** the four bonds of methane point toward the corners of a regular tetrahedron; the ideal $H-C-H$ angle is about 109°28′ (109.5°).



1.2 Hybridisation in Organic Compounds

- Hybridisation is the mixing of atomic orbitals of comparable energies on the same atom to produce new orbitals oriented for bonding.

- The number of hybrid orbitals formed equals the number of atomic orbitals mixed.



1.2.1 Types of Carbon Hybridisation

Property	sp^3	sp^2	sp
Orbitals mixed	1 s + 3 p	1 s + 2 p	1 s + 1 p
Hybrid orbitals formed	4	3	2
Unhybridised p orbitals	0	1	2
Typical bonding	4 σ	3 σ + 1 π	2 σ + 2 π
Ideal geometry	Tetrahedral	Trigonal planar	Linear
Bond angle	109.5°	120°	180°
s-character	25%	33⅓%	50%
Typical example	CH_4 / ethane	Ethene (C_2H_4)	Ethyne (C_2H_2)

Note: A π bond is formed by sideways overlap of unhybridised p orbitals; σ bonds arise from end-on overlap. A double bond consists of one σ and one π bond; a triple bond consists of one σ and two π bonds.

1.3 Determination of Hybridisation

1.3.1 Method I — Number of π Bonds

For a neutral carbon in ordinary hydrocarbons with four bond-order units, the following quick rule is useful:

Number of π bonds at carbon	0	1	2
Hybridisation	sp^3	sp^2	sp

- $CH_3 - CH_3$: both carbons are sp^3 hybridised.
- $CH_2 = CH_2$: both carbons are sp^2 hybridised.
- $HC \equiv CH$: both carbons are sp hybridised.
- In benzene, every ring carbon is sp^2 hybridised; in diamond, carbon is sp^3; in graphite, each carbon in a sheet is sp^2.

Important limitation: Counting π bonds alone is not sufficient for every charged or resonance-stabilised species; consider electron domains and conjugation as well.

1.3.2 Method II — Electron-Pair / Steric-Number Method

Count the σ-bonded atoms attached to the central atom and the lone pairs localised on that atom. A multiple bond contributes one electron domain for this purpose.

$$Steric\ number\ (SN) = number\ of\ \sigma\ bonds + number\ of\ lone\ pairs$$

Steric number	2	3	4
Common hybridisation	sp	sp^2	sp^3
Electron-domain arrangement	Linear	Trigonal planar	Tetrahedral

- Methane, CH_4: 4 σ bonds + 0 lone pairs → SN = 4 → sp^3.
- Ethene, $CH_2 = CH_2$: each carbon has 3 σ bonds + 0 lone pairs → SN = 3 → sp^2.
- Ethyne, $HC \equiv CH$: each carbon has 2 σ bonds + 0 lone pairs → SN = 2 → sp.
- Carbocation CH_3^+: three σ bonds and an empty p orbital → usually trigonal planar, sp^2.
- Carbanion CH_3^-: three σ bonds and one lone pair → commonly pyramidal, approximately sp^3; conjugation may favour sp^2.
- Carbon radical CH_3 •: often approximately trigonal planar with a p-type singly occupied orbital; the precise geometry depends on substitution and environment.

For elements beyond the second period, simple steric-number labels such as sp^3d and sp^3d^2 are sometimes used in introductory models; these are not needed for the carbon examples in this chapter.

1.4 Applications of Hybridisation

1.4.1 Size and s-Character of Hybrid Orbitals

An orbital with greater s-character is, on average, more concentrated near the nucleus. For carbon hybrid orbitals, the qualitative order of radial extent is:

$$Size: sp^3 > sp^2 > sp$$

$$s-character: sp > sp^2 > sp^3$$

1.4.2 Electronegativity of Hybridised Carbon

Greater s-character holds bonding electrons closer to the carbon nucleus.

- Its electrons are held more strongly by the carbon nucleus.
- The hybrid orbital is generally more contracted.
- Carbon attracts the shared bonding electrons more strongly.

Hence the effective electronegativity of carbon in comparable bonds generally increases in the order:

$$sp > sp^2 > sp^3$$

1.4.2.1 Effect of Hybridisation on C–H Bond Polarisation

This trend helps explain the stronger polarisation of $C-H$

- Consider a carbon–hydrogen (C–H) bond.
- When carbon and hydrogen form a covalent bond, they share a pair of electrons. However, the shared electrons are not necessarily distributed equally.
- If carbon attracts the shared electrons more strongly than hydrogen, the electron density shifts towards carbon.

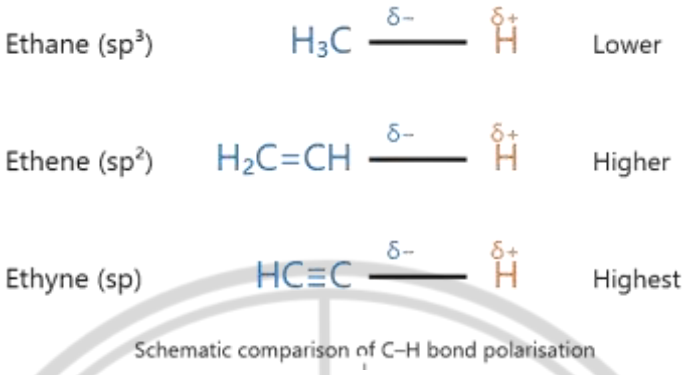
Consequently:

- Carbon develops a partial negative charge (δ−).
- Hydrogen develops a partial positive charge (δ+).
- The C–H bond becomes polarised.

Now compare three hydrocarbons.

Ethane, ethene and ethyne

$$\boxed{C(sp^3)-H < C(sp^2)-H < C(sp)-H}$$



Schematic comparison of C–H bond polarisation

In comparable situations, the carbon atom in ethyne has greater effective electronegativity because it is sp-hybridised.

Therefore, the general tendency for C–H bond polarisation is:

$$\boxed{C(sp^3) - H < C(sp^2) - H < C(sp) - H}$$

1.4.2.2 Effect of Hybridisation on Acidity of C–H Bonds

Terminal alkynes more acidic

Consider three compounds:

1. Ethane: CH_3–CH_3 (sp^3 carbon)
2. Ethene: CH_2=CH_2 (sp^2 carbon)
3. Ethyne: HC≡CH (sp carbon)

When a hydrogen atom is removed as a proton (H^+), the remaining carbon species carries a negative charge.

For ethyne:

$$HC \equiv CH \rightleftharpoons HC \equiv C^- + H^+$$

- The negatively charged carbon is sp-hybridised.
- Because sp-hybridised carbon has 50% s-character, it holds the negative charge closer to the nucleus and stabilises it more effectively.
- In comparison, an sp^3-hybridised carbon has only 25% s-character and is less effective at stabilising a negative charge.
- Important principle: The more stable the conjugate base formed after losing H^+, the stronger the acid.

However, hybridisation is not the only factor that determines the stability of a negatively charged carbon atom or the acidity of a C–H bond. Formal charge, electron-withdrawing substituents and resonance can significantly modify the expected trend.

(i) Effect of Resonance

Example: Ethyne and cyclopentadiene

Ethyne (HC≡CH) contains sp-hybridised carbon, whereas cyclopentadiene (C_5H_6) contains an sp³-hybridised CH_2 carbon.

According to hybridisation alone, we might expect ethyne to be more acidic.

However:

Compound	Approximate pKa
Ethyne	25
Cyclopentadiene	16

Explanation:

When cyclopentadiene loses H^+ from its CH_2 group, it forms the cyclopentadienyl anion ($C_5H_5^-$).

This anion is exceptionally stable because:

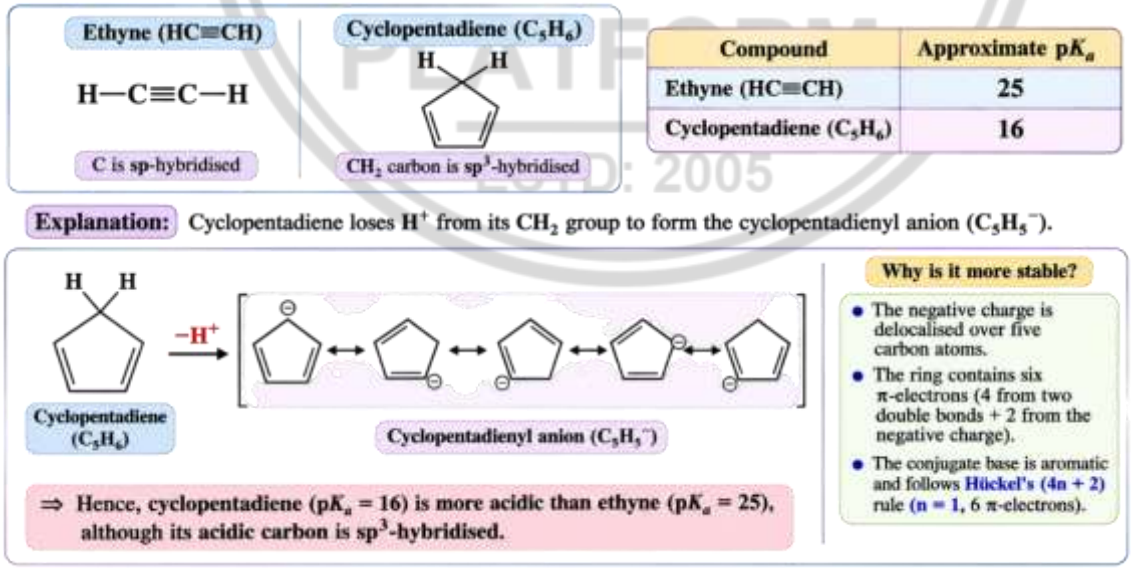
- The negative charge is delocalised over five carbon atoms.
- The ring contains six π-electrons.
- The conjugate base is aromatic and follows Hückel's $4n + 2$rule.

In contrast, the acetylide ion ($HC{\equiv}C^-$) is stabilised by the high s-character of its sp-hybridised carbon, but it does not have comparable aromatic resonance stabilisation.

Therefore,

Cyclopentadiene > Ethyne

Conclusion: Resonance and aromatic stabilisation can outweigh the effect of hybridisation.



(ii) Effect of Electron-Withdrawing Substituents (−I Effect)

Example: Methane and nitromethane

Compare:

$$CH_4 \quad \text{and} \quad CH_3NO_2$$

- Both compounds contain an sp³-hybridised carbon atom.
- However, their acidities differ considerably.

Compound	Approximate pKa
Methane	50
Nitromethane	10 (aqueous)

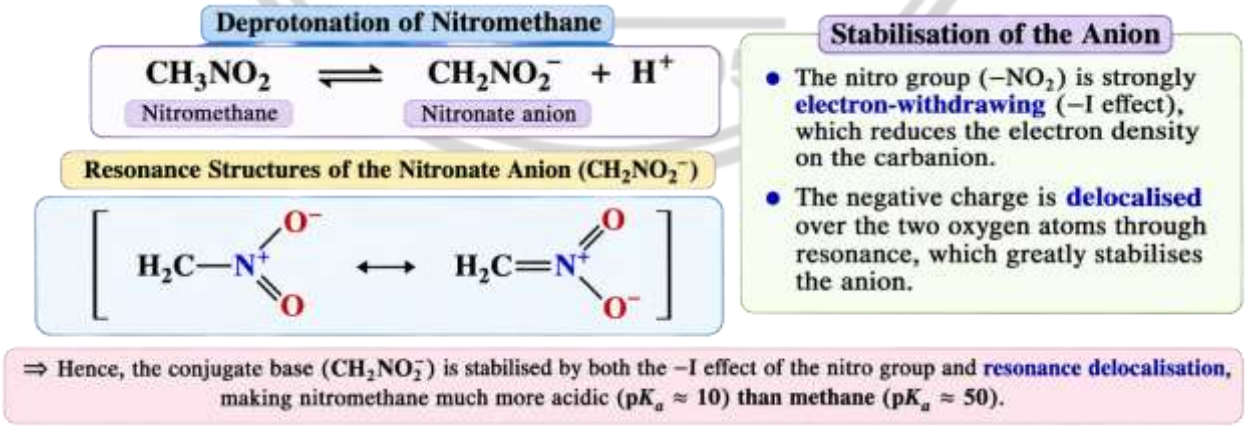
- The nitro group ($-NO_2$) is strongly electron-withdrawing.
- When nitromethane loses H^+, it forms a nitronate anion:

$$CH_3NO_2 \rightleftharpoons CH_2NO_2^- + H^+$$

The negative charge is stabilised by the electron-withdrawing effect of the nitro group and, importantly, by resonance delocalisation involving the oxygen atoms.

Therefore,

Nitromethane > Methane



Conclusion: Electron-withdrawing substituents can greatly increase acidity even when the original C–H carbon has the same hybridisation.

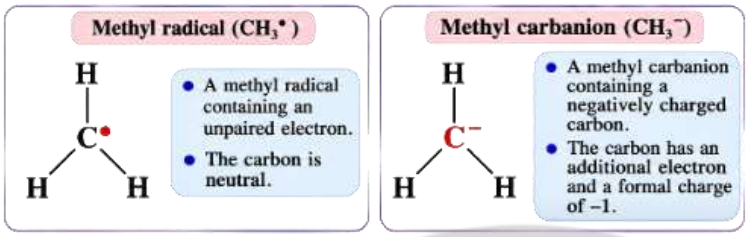
(iii) Effect of Formal Charge

Example: Neutral carbon and negatively charged carbon

Consider the following species:

$$CH_3^{\cdot} \quad \text{and} \quad CH_3^-$$

- CH_3• is a methyl radical containing an unpaired electron.
- CH_3^- is a methyl carbanion containing a negatively charged carbon.
- In CH_3^-, the carbon atom has an additional electron and a formal charge of −1.



- This negative charge changes the electron distribution and makes the carbon centre strongly electron-rich.
- Consequently, CH_3^- behaves as a strong base and nucleophile.

Important: The electronegativity trend for neutral hybridised carbon cannot be applied directly to charged species without considering their electronic structures. Formal charge also affects basicity, nucleophilicity and stability; it does not, by itself, establish an acidity order.

1.4.3 Bond-Length Variation

As bond order increases, $C-C$ bonds generally become shorter. Greater s-character also tends to shorten comparable $C-H$ bonds.

Property	Ethane	Ethene	Ethyne
Representative **C – H** bond length	About 1.09 Å	About 1.09 Å	About 1.06 Å
Representative carbon–carbon length	About 1.54 Å single bond	About 1.34 Å double bond	About 1.20 Å triple bond

$$C-C\ bond\ length: single > double > triple$$

- Rounded teaching values for the named molecules; exact bond lengths depend on the structural definition and measurement. Use the qualitative trend rather than extra decimal places.

1.4.4 Bond Strength in Hydrocarbons

- In otherwise comparable bonds, shorter bonds are often stronger because of more effective orbital overlap.
- Bond strength also depends on bond order and molecular environment.

Comparison	$C-C$ single bond	$C=C$ double bond	$C\equiv C$ triple bond
Total bond order	One	Two	Three
Qualitative total bond strength in comparable hydrocarbon examples	Lower	Intermediate	Higher

- Bond dissociation enthalpy is defined for a specified bond-breaking process and physical state. Average bond enthalpies are different quantities.
- A double bond is not twice as strong as a single bond, and a triple bond is not three times as strong. Compare named molecules and consistent data definitions.

2 Steric effects

2.1 Definition of Steric Effect

The steric effect is the phenomenon in which the presence of bulky atoms or groups near a reaction centre creates spatial obstruction, making it difficult for an attacking reagent to approach the reaction site.

As a result, the reaction may become slower or may follow a different reaction mechanism. This phenomenon is called steric hindrance.

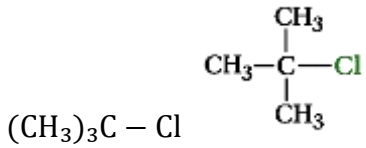
Important points:

- Steric hindrance arises due to the large size and spatial arrangement of atoms or groups.
- Bulky groups near the reaction centre obstruct the approach of attacking reagents.
- Steric hindrance particularly affects substitution, elimination and resonance.
- It may decrease the rate of a reaction or favour an alternative reaction pathway.

2.2 Effect of Steric Hindrance on Alkyl Halides

(i) Tertiary Alkyl Halides

Tertiary alkyl halides contain three alkyl groups attached to the carbon atom bearing the halogen. For example, tert-butyl chloride:



$(CH_3)_3C-Cl$

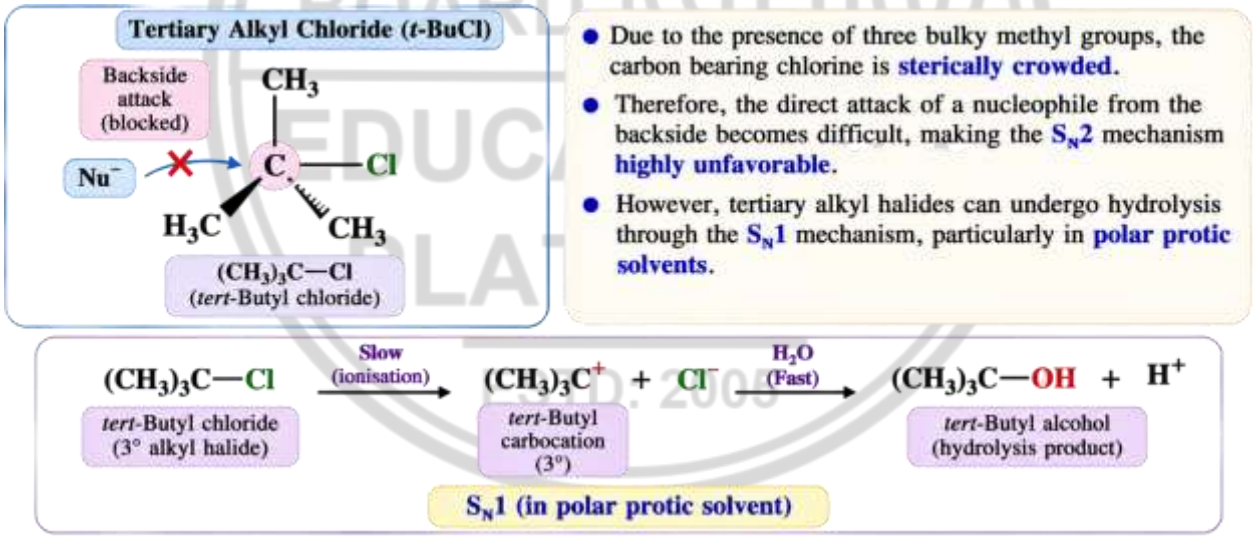
- Due to the presence of three bulky methyl groups, the carbon bearing chlorine is sterically crowded.
- Therefore, the direct attack of a nucleophile from the backside becomes difficult, making the S_n2 mechanism highly unfavorable.
- However, tertiary alkyl halides can undergo hydrolysis through the S_n1 mechanism, particularly in polar protic solvents.
- Formation of carbocation:

$$(CH_3)_3C-Cl \rightarrow (CH_3)_3C^+ + Cl^-$$

- The tertiary carbocation is relatively stable because of the +I effect and hyperconjugation of the three methyl groups.
- The carbocation then reacts with water to form an alcohol.

$$(CH_3)_3C^+ + H_2O \rightarrow (CH_3)_3COH_2^+$$

$$(CH_3)_3COH_2^+ \rightarrow (CH_3)_3COH + H^+$$



Conclusion: Tertiary alkyl halides generally favour the S_n1 mechanism over S_n2 because of steric hindrance and the relative stability of tertiary carbocations.

2.3 Steric Hindrance in Primary Alkyl Halides Containing a Quaternary β-Carbon

Consider neopentyl chloride:

$$(CH_3)_3C - CH_2 - Cl$$

Although neopentyl chloride is a primary alkyl halide, it undergoes nucleophilic substitution very slowly.

Reason:

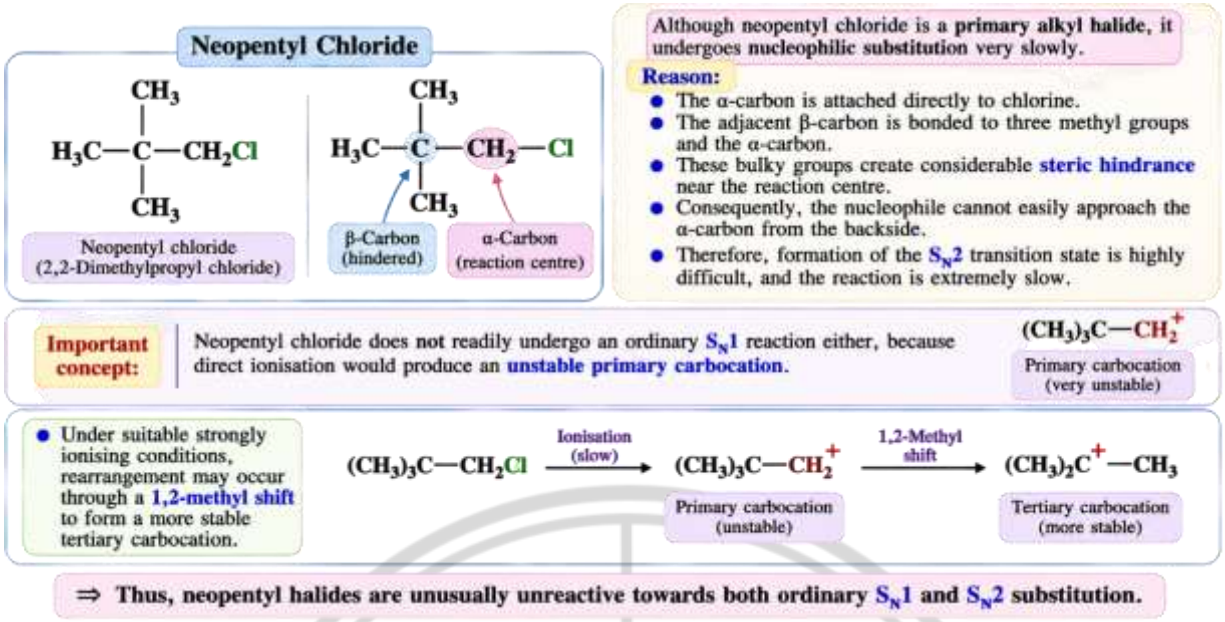
- The α-carbon is attached directly to chlorine.
- The adjacent β-carbon is bonded to three methyl groups and the α-carbon.
- These bulky groups create considerable steric hindrance near the reaction centre.
- Consequently, the nucleophile cannot easily approach the α-carbon from the backside.

Therefore, formation of the S_n2 transition state is highly difficult, and the reaction is extremely slow.

Important concept: Neopentyl chloride does not readily undergo an ordinary S_n1 reaction either, because direct ionisation would produce an unstable primary carbocation.

$$(CH_3)_3C - CH_2^+$$

- Under suitable strongly ionising conditions (Lewis acid such as Ag^+, Strongly polar, weakly nucleophilic solvents (HCOOH), (HFIP), (CF_3COOH), rearrangement may occur through a 1,2-methyl shift to form a more stable tertiary carbocation.
- However, this does not mean that neopentyl chloride readily forms a carbocation during normal hydrolysis.
- Thus, neopentyl halides are unusually unreactive towards both ordinary S_n1 and S_n2 substitution.



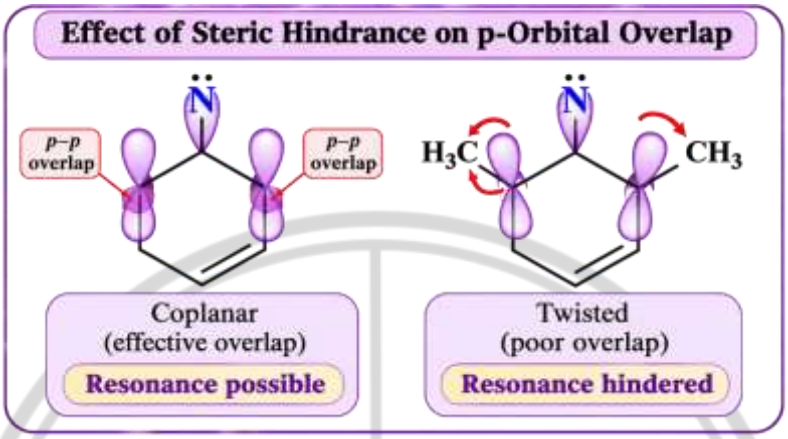
2.4 Steric Inhibition of Resonance (SIR)

Definition: Steric inhibition of resonance is the phenomenon in which bulky groups prevent two conjugated groups from attaining the coplanar arrangement required for effective orbital overlap, thereby decreasing or preventing resonance.

- For resonance to occur effectively, the participating p-orbitals must be properly aligned.
- When bulky groups are present near the conjugated system, steric repulsion may force the groups to twist out of the same plane.
- As a result, overlap between the p-orbitals decreases and resonance becomes weaker.

Example: 2,6-Dimethyl-N,N-dimethylaniline

- In aniline derivatives, the lone pair of electrons on nitrogen can participate in resonance with the benzene ring.
- However, when bulky methyl groups are present at both ortho positions, they can interfere with the alignment of the nitrogen lone pair with the benzene π-system.
- Aniline: Lone-pair conjugation with the ring.
- 2,6-Dimethyl-N,N-dimethylaniline: Ortho methyl groups hinder effective conjugation.
- Due to steric crowding, the nitrogen-containing group may twist relative to the benzene ring.



This reduces the overlap between the nitrogen lone pair and the π-electron system. Consequently, the resonance interaction becomes weaker.

Important consequences of SIR

- Reduced delocalisation of electrons.
- Decreased resonance stabilisation.
- Changes in basicity and other chemical properties.
- Reduced conjugation between adjacent groups.

3 Inductive effect and its applications

3.1 Bond polarity and electronegativity

When two bonded atoms attract electrons unequally, the bonding electron density is displaced towards the more electronegative atom.

- The bond is polar even though it remains covalent.
- In CH_3Cl the carbon bonded to chlorine is electron deficient relative to a comparable hydrocarbon carbon, while chlorine carries a partial negative charge.
- Partial charges δ+ and δ− describe unequal sharing; they are not full ionic charges.
- A bond dipole is a vector. Molecular dipole moment depends on both bond polarities and geometry.
- A molecule may contain polar bonds yet have zero net dipole moment when symmetry cancels their contributions, as in CCl_4.

- When two bonded atoms attract electrons **unequally**, the bonding electron density is displaced towards the **more electronegative atom**.
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- Partial charges δ^+ and δ^- describe unequal sharing; they are not full ionic charges.
- A bond dipole is a **vector**.
- Molecular dipole moment depends on both **bond polarities** and **geometry**.
- A molecule may contain polar bonds yet have zero net dipole moment when symmetry cancels their contributions, as in CO_2.

Polar Bond (Example: CH_3Cl)

$\delta^+ \quad \delta^-$
$CH_3 — Cl$

Polar Bonds but Zero Net Dipole (Example: CO_2)

$\delta^- \quad \delta^+ \quad \delta^-$
$O = C = O$

Net dipole moment = 0

3.2 Definition and transmission of the inductive effect

Inductive Effect (I-Effect)

Definition

The inductive effect is the permanent polarisation of σ-bonds caused by an electron-withdrawing or electron-releasing atom or group, resulting in a change in electron distribution along the molecular framework.

Important Characteristics of the Inductive Effect

1. Permanent effect: The inductive effect is a permanent electronic effect arising from differences in the electron-attracting tendencies of atoms or groups.

2. Transmission through σ-bonds: In the conventional model, the inductive effect is transmitted through σ-bonds connecting atoms in a molecule.

3. Decreases with distance: The effect is strongest near the substituent and decreases rapidly as the number of intervening σ-bonds increases.

For example, in 1-chloropropane:

$$Cl - CH_2 - CH_2 - CH_3$$

Chlorine withdraws electron density through its −I effect. Its influence decreases progressively along the carbon chain.

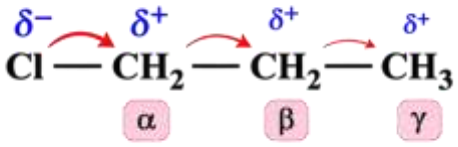
$$\boxed{\alpha\text{-carbon} > \beta\text{-carbon} > \gamma\text{-carbon}}$$

- However, the effect becomes progressively **weaker** with increasing **distance** from chlorine.
- Therefore,

Inductive effect decreases with distance.

- The effect is generally **strongest** at the carbon directly attached to the electron-withdrawing group and becomes much weaker after several σ-bonds.

Example : 1-Chloropropane



Inductive effect (–I) of Cl :
α-carbon > β-carbon > γ-carbon



4. No π-system required: The inductive effect operates through σ-bonds and does not require double bonds, triple bonds or a conjugated π-system.

5. No attacking reagent required: The inductive effect exists in the molecule even in the absence of an attacking reagent.

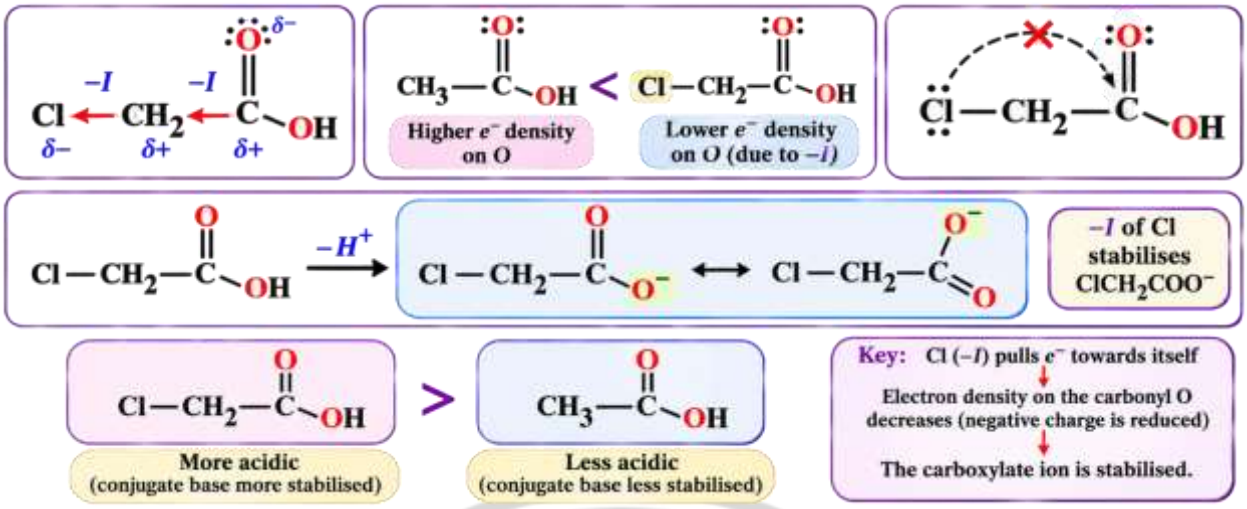
6. Influence on nearby functional groups: An electron-withdrawing substituent can influence a nearby functional group without being directly conjugated with it.

Example: Chloroacetic acid

$$Cl - CH_2 - COOH$$

- Chlorine withdraws electron density through its −I effect.
- This electron-withdrawing influence reaches the nearby carboxyl group through the σ-bond framework.
- The chlorine atom is separated from the carboxyl group by a saturated CH_2 carbon; therefore, direct resonance conjugation between chlorine and the carboxyl group is absent.
- The electron-withdrawing -I effect helps stabilise the negatively charged conjugate base, $ClCH_2COO^-$ after the loss of H^+.
- Therefore, chloroacetic acid is more acidic than acetic acid.

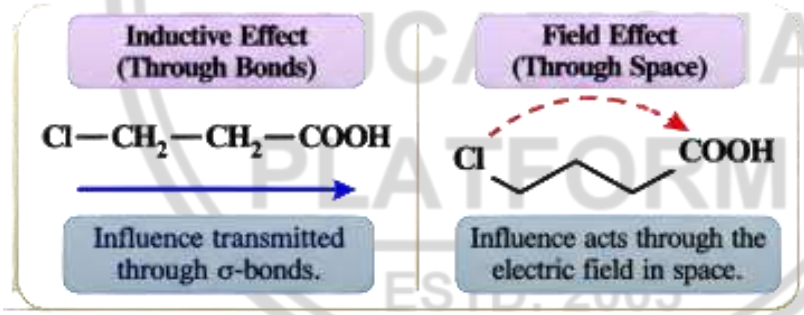
$$\boxed{ClCH_2COOH > CH_3COOH}$$



(Order of acidity)

7. Through-space field effect: Modern physical organic chemistry also recognises that a polar substituent can influence another part of a molecule through its electric field, without the influence necessarily being transmitted along σ-bonds.

- Inductive effect: Electron-withdrawing or electron-releasing influence transmitted through the σ-bond framework.
- Field effect: Electrostatic influence exerted directly through space.



- The field effect depends on the spatial distance and orientation of the interacting groups.
- In actual molecules, both through-bond and through-space contributions may operate simultaneously.

8. Effect of Number of Electron-Withdrawing Substituents

When more electron-withdrawing groups are attached near the same reaction centre, their combined electron-withdrawing influence generally increases.

Important points:

1. Each electron-withdrawing substituent contributes to the overall −I effect.
2. Increasing the number of such substituents generally increases the electron-withdrawing influence.
3. In carboxylic acids, this can increase the stability of the carboxylate conjugate base.
4. Consequently, acidity generally increases with the number of electron-withdrawing substituents under comparable conditions.

Hence, the order of acidity is:

$$\boxed{CCl_3COOH > CHCl_2COOH > CH_2ClCOOH > CH_3COOH}$$

Or,

$$\boxed{\begin{array}{c}\text{Trichloroacetic acid} \\ > \text{Dichloroacetic acid} \\ > \text{Chloroacetic acid} \\ > \text{Acetic acid}\end{array}}$$

3.3 Electron-Withdrawing Groups and the Negative Inductive Effect (−I Effect)

Definition

The negative inductive effect (−I effect) is the permanent electron-withdrawing effect produced by an atom or group that attracts electron density through the σ-bond framework relative to hydrogen in a comparable molecular environment.

Important Characteristics of the −I Effect

1. Electron-withdrawing nature: Groups exhibiting the −I effect attract electron density towards themselves through σ-bonds, making the neighbouring atoms relatively electron-deficient.

2. Common electron-withdrawing groups: Some important groups showing the −I effect are:

- Positively charged ammonium groups ($-NR_3^+$)
- Nitro group ($-NO_2$)
- Cyano group (−CN)
- Carbonyl-containing groups (such as −CHO, −COR and −COOH)
- Halogens (−F, −Cl, −Br and −I)

3. Dependence on molecular environment: The strength of the −I effect depends on the nature of the electron-withdrawing group, its point of attachment and the surrounding molecular environment.

3.3.1 Negative Inductive Effect of Halogens

Halogens are more electronegative than carbon and therefore withdraw electron density through σ-bonds.

For halogens attached through the same type of bond, the usual qualitative order of the −I effect is:

$$\boxed{-F > -Cl > -Br > -I}$$

Explanation:

- Fluorine is the most electronegative halogen and generally exhibits the strongest −I effect.

- Chlorine has a weaker −I effect than fluorine.
- Bromine has a weaker −I effect than chlorine.
- Iodine generally exhibits the weakest −I effect among these halogens.

Important: This order represents the relative electron-withdrawing inductive effects of halogens under comparable bonding conditions. It should not be used as a universal order for acidity, basicity, reactivity or other chemical properties.

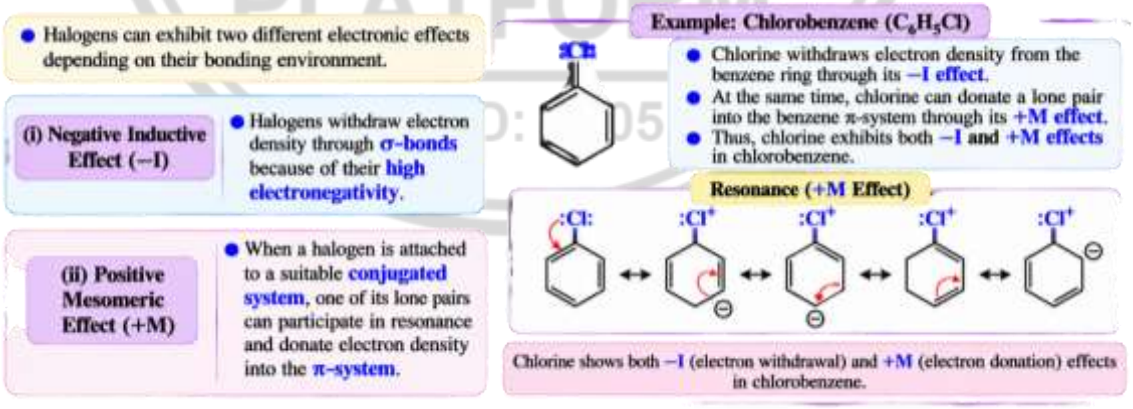
3.3.2 Inductive Effect and Resonance Effect of Halogens

Halogens can exhibit two different electronic effects depending on their bonding environment.

1. Negative inductive effect (−I): Halogens withdraw electron density through σ-bonds because of their high electronegativity.

2. Positive mesomeric effect (+M): When a halogen is attached to a suitable conjugated system, one of its lone pairs can participate in resonance and donate electron density into the π-system.

Example 1 : Chlorobenzene (C_6H_5Cl)



Example 2. Vinyl Chloride (Chloroethene)

$$\boxed{CH_2 = CH - Cl}$$

Vinyl chloride is an important example of a halogen attached directly to an alkene carbon.

Explanation:

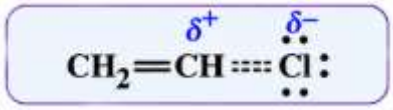
- Chlorine is directly attached to an sp^2-hybridised carbon.
- Chlorine possesses three lone pairs of electrons.
- A suitably oriented lone pair on chlorine can overlap with the adjacent π-system.
- Consequently, chlorine can donate electron density through resonance (+M effect).
- At the same time, chlorine withdraws electron density through the σ-bond (−I effect).

The resonance can be represented as:

$$CH_2 = CH - Cl \leftrightarrow {}^{-}CH_2 - CH = Cl^{+}$$

- The charge-separated structure is a minor resonance contributor.
- Thus, vinyl chloride exhibits both −I and +M effects.

Result : Due to resonance, the C–Cl bond acquires partial double-bond character, making it shorter and generally stronger than an ordinary C(sp^3)–Cl bond.



3.4 Electron-Releasing Groups and the Positive Inductive Effect (+I Effect)

Definition

The positive inductive effect (+I effect) is the permanent electron-releasing effect produced by an atom or group that pushes electron density towards the attached molecular framework through σ-bonds, relative to hydrogen in a comparable molecular environment.

Important Characteristics of the +I Effect

1. Electron-releasing nature: Groups exhibiting the +I effect increase electron density in the neighbouring molecular framework through σ-bond polarisation.

2. Permanent effect: The +I effect is a permanent electronic effect and does not require an attacking reagent.

3. Transmission through σ-bonds: In the conventional model, the +I effect is transmitted through σ-bonds and decreases rapidly with increasing bond distance.

4. Common +I groups: Alkyl groups such as $-CH_3$, $-C_2H_5$, $-CH(CH_3)_2$ and $-C(CH_3)_3$ are commonly treated as electron-releasing groups.

5. Relative to hydrogen: The electron-releasing tendency is conventionally compared with hydrogen, which is taken as the reference group (hydrogen is assigned a zero inductive effect for comparison purposes).

3.4.1 Order of +I Effect of Alkyl Groups

The conventional qualitative order of the +I effect of alkyl groups is:

$$\boxed{-\mathbf{C(CH_3)_3} > -\mathbf{CH(CH_3)_2} > -\mathbf{CH_2CH_3} > -\mathbf{CH_3} > -\mathbf{H}}$$

Or,

$$\boxed{\text{tert-Butyl} > \text{Isopropyl} > \text{Ethyl} > \text{Methyl} > \text{Hydrogen}}$$

Explanation:

- tert-Butyl group: Conventionally shows the strongest +I effect among the listed alkyl groups.
- Isopropyl group: Shows a weaker +I effect than tert-butyl but a stronger effect than ethyl.
- Ethyl group: Shows a stronger +I effect than methyl in the conventional comparison.
- Methyl group: Shows a relatively weak +I effect.
- Hydrogen: Taken as the reference for comparing electron-releasing and electron-withdrawing effects.
- Thus, in the elementary model, increasing alkyl substitution generally increases the electron-releasing inductive influence.

3.4.2 The +I Effect of Alkyl Groups Not a Universal Rule

The conventional order of the +I effect is:

$$\boxed{\mathbf{3^\circ} > \mathbf{2^\circ} > \mathbf{1^\circ} > \text{Methyl} > \boldsymbol{H}}$$

- This means that, in the elementary model, a tertiary alkyl group has a stronger electron-releasing inductive influence than a secondary or primary alkyl group.

- But does this mean that a compound containing a tertiary alkyl group will always be more stable, more reactive or more basic?
- No. The +I effect is only one factor that influences the properties of organic compounds.
- Other factors, such as hyperconjugation, steric hindrance, solvation and molecular geometry, may reinforce or oppose its influence.
- Let us understand this with examples.

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(i) Hyperconjugation — Why +I Effect Alone Cannot Explain Stability

- Consider three carbocations:

$\mathbf{CH_3CH_2^+}$	**Primary carbocation**
$\mathbf{(CH_3)_2CH^+}$	**Secondary carbocation**
$\mathbf{(CH_3)_3C^+}$	**Tertiary carbocation**

- The conventional stability order of simple alkyl carbocations is:

$$\boxed{\mathbf{3^\circ > 2^\circ > 1^\circ}}$$

Why is the tertiary carbocation generally more stable?

There are two important reasons.

A. Positive inductive effect (+I)

- Alkyl groups release electron density towards the positively charged carbon.
- This helps reduce its electron deficiency.
- A tertiary carbocation has three alkyl groups attached to the positively charged carbon.
- Therefore, it receives a greater electron-releasing inductive influence in the conventional model.

B. Hyperconjugation

- Hyperconjugation is the delocalisation of electrons from a suitably aligned σ-bond into an adjacent vacant p-orbital or π-system.
- In a carbocation, the positively charged carbon has an empty p-orbital.
- An adjacent C–H σ-bond can overlap with this empty p-orbital, allowing electron density to become delocalised.
- For simple methyl-substituted carbocations:

Carbocation	Number of adjacent C–H bonds available for hyperconjugation
$CH_3CH_2^+$	3
$(CH_3)_2CH^+$	6
$(CH_3)_3C^+$	9

A greater number of suitable hyper-conjugative interactions generally provides greater stabilisation.

Therefore, the tertiary carbocation is stabilised by both the +I effect and hyperconjugation.

Conclusion: We should not explain carbocation stability using the +I effect alone.

(ii) Steric Effect — Why a Stronger +I Effect Does Not Always Mean Faster Reaction

Consider two alkyl chlorides.

Ethyl chloride (Primary alkyl halide)

$$\mathbf{CH_3CH_2Cl}$$

The carbon bearing chlorine is relatively accessible to an attacking nucleophile.

tert-Butyl chloride (Tertiary alkyl halide)

$$\mathbf{(CH_3)_3CCl}$$

The carbon bearing chlorine is surrounded by three bulky methyl groups.

In an S_n2 reaction, the nucleophile must approach the carbon bearing chlorine from the side opposite the C–Cl bond.

What happens in ethyl chloride?

- The reaction centre is relatively less crowded.
- The nucleophile can approach from the backside.

- Therefore, an S_n2 reaction can occur comparatively easily under suitable conditions.

What happens in tert-butyl chloride?

- Three methyl groups surround the carbon bearing chlorine.
- These groups obstruct the backside approach of the nucleophile.
- Formation of the S_n2 transition state becomes extremely difficult.
- Consequently, tert-butyl chloride is highly unfavourable for S_n2 substitution.

For comparable simple alkyl halides, the general S_n2 reactivity order is:

$$\text{Methyl} > 1^\circ > 2^\circ \gg 3^\circ$$

Notice the difference:

- The conventional +I effect increases with alkyl substitution.
- S_n2 reactivity generally decreases with increasing steric crowding.

Conclusion: Although tert-butyl has a stronger conventional +I effect than ethyl, tert-butyl chloride undergoes S_n2 substitution much more slowly because steric hindrance dominates.

This does not mean tert-butyl chloride is unreactive in every reaction; it can undergo S_n1 substitution under suitable conditions.

(iii) Solvation — How a Solvent Can Change Acidity or Basicity

Solvation is the process by which solvent molecules surround and stabilise dissolved molecules or ions. When water acts as the solvent, this process is called **hydration**.

In aqueous solution, the basicity of amines depends not only on the electron-releasing inductive effect (+I) of alkyl groups but also on the stabilisation of the protonated amine by water molecules.

1. Effect of the +I Effect on Basicity

Consider methylamine and dimethylamine:

- **Methylamine:** CH_3NH_2

- **Dimethylamine:** $(CH_3)_2NH$

Both contain electron-releasing methyl groups.

According to the +I effect:

1. Dimethylamine contains two methyl groups, whereas methylamine contains only one.
2. Methyl groups donate electron density towards nitrogen through the +I effect.
3. Therefore, dimethylamine generally has greater electron density available at nitrogen.
4. This favours the acceptance of H^+ by dimethylamine.

Thus, based on the +I effect alone:

Dimethylamine > Methylamine

2. Effect of Solvation on Basicity

- When an amine accepts a proton (H^+), it forms a positively charged conjugate acid.

Methylamine:

$CH_3NH_2 + H^+ \rightleftharpoons CH_3NH_3^+$

Dimethylamine:

$(CH_3)_2NH + H^+ \rightleftharpoons (CH_3)_2NH_2^+$

- The positively charged ammonium ions are stabilised by surrounding water molecules through hydrogen bonding and ion–dipole interactions.

Explanation:

1. Water molecules surround the protonated amine and stabilise its positive charge.
2. The oxygen atom of water can form hydrogen bonds with the N–H bonds of the protonated amine.
3. Methylammonium ion ($CH_3NH_3^+$) has three N–H bonds and can form favourable hydrogen bonds with water.
4. Dimethylammonium ion ($(CH_3)_2NH_2^+$) has two N–H bonds and can also undergo effective hydration.

5. Although methylammonium generally has greater hydrogen-bonding accessibility, the stronger +I effect of two methyl groups in dimethylamine contributes sufficiently to make dimethylamine more basic overall.

3. Example: Dimethylamine and Trimethylamine

Consider:

- **Dimethylamine:** $(CH_3)_2NH$ — secondary amine
- **Trimethylamine:** $(CH_3)_3N$ — tertiary amine

According to the +I effect alone, trimethylamine might be expected to be more basic because it contains three electron-releasing methyl groups.

However, in aqueous solution, dimethylamine is generally more basic than trimethylamine at room temperature.

Explanation:

1. Trimethylamine has a stronger electron-releasing inductive influence due to its three methyl groups.
2. However, its protonated form, $(CH_3)_3NH^+$, is relatively sterically crowded and has only one N–H bond.
3. The protonated form of dimethylamine, $(CH_3)_2NH_2^+$, has two N–H bonds and is generally more accessible to water molecules.
4. Therefore, the conjugate acid of dimethylamine can be hydrated more effectively.
5. This favourable hydration contributes to dimethylamine being more basic than trimethylamine in water.

Thus, the basicity order is:

Dimethylamine > Trimethylamine

4. Order of Basicity of Simple Methylamines in Water

At approximately 25°C, the commonly observed order is:

$(CH_3)_2NH > CH_3NH_2 > (CH_3)_3N > NH_3$

This order results from the combined influence of:

1. **+I effect:** Alkyl groups increase electron density on nitrogen.

2. **Solvation:** Water stabilises neutral and protonated amines to different extents.
3. **Steric effects:** Bulky alkyl groups can hinder effective hydration.

Conclusion

- A stronger +I effect does not necessarily produce a stronger base in every solvent.
- In aqueous solution, **the basicity of an amine depends on the combined effects of electron donation, solvation and steric hindrance**.

(iv) Molecular Geometry — Why Orbital Alignment Matters

Consider hyperconjugation again.

For effective hyperconjugation, a suitable σ-bond must be properly aligned with an adjacent p-orbital or π-system.

- Favourable orbital alignment allows effective overlap.
- Factor effecting the alignment:

  Rotation around a C–C single bond,

  Molecular rigidity: In some cyclic or rigid molecules, rotation around certain bonds is restricted.

  Steric hindrance: Bulky groups may favour particular molecular conformations, which can influence orbital alignment.
- Poor orbital alignment reduces overlap.
- Restricted molecular geometry may therefore weaken hyperconjugation even when electron-releasing alkyl groups are present.

Consequently, two molecules with similar alkyl substitution may exhibit different stability or reactivity because of differences in molecular geometry.

4 Resonance and the mesomeric effect

4.1 Introduction

In organic compounds, electrons present in covalent bonds may undergo displacement due to the influence of neighbouring atoms or groups.

- In certain molecules, particularly those containing conjugated double bonds or lone pairs of electrons adjacent to multiple bonds, the electrons are not confined to a particular bond or atom. Instead, they become delocalized over two or more adjacent atoms.

This phenomenon is known as resonance or mesomerism.

4.2 Important Points:

1. Resonance involves the delocalization of electrons within a molecule or ion.

2. Only π-electrons, lone pairs of electrons, and, in suitable systems, electrons in certain σ-bonds can participate in resonance.

3. The positions of the nuclei remain unchanged during resonance.

4. Resonance does not involve the actual movement of atoms from one position to another.

5. The actual molecule is represented by a resonance hybrid of contributing canonical structures.

6. Electron delocalization generally increases the stability of a molecule or ion.

4.3 Conditions Necessary for Mesomeric Effect

For the mesomeric effect to operate effectively, the molecule must possess a suitable conjugated system.

(i) Presence of a Conjugated System

A conjugated system contains adjacent atoms having p-orbitals that can overlap continuously.

For example:

$$CH_2 = CH - CH = CH_2$$

- This compound is 1,3-butadiene.
- The two double bonds are separated by one single bond, allowing continuous overlap of p-orbitals.
- The π-electrons are delocalized over the four carbon atoms rather than being completely localized between individual pairs of carbon atoms.

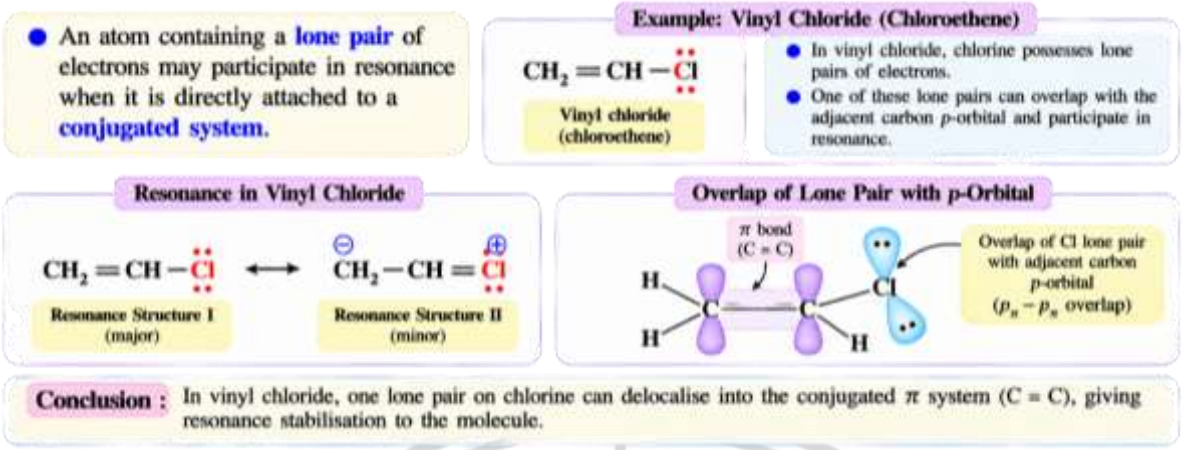
(ii) Presence of Lone Pairs Adjacent to a π-Bond

An atom containing a lone pair of electrons may participate in resonance when it is directly attached to a conjugated system.

For example: $CH_2 = CH - Cl$

In vinyl chloride, chlorine possesses lone pairs of electrons.

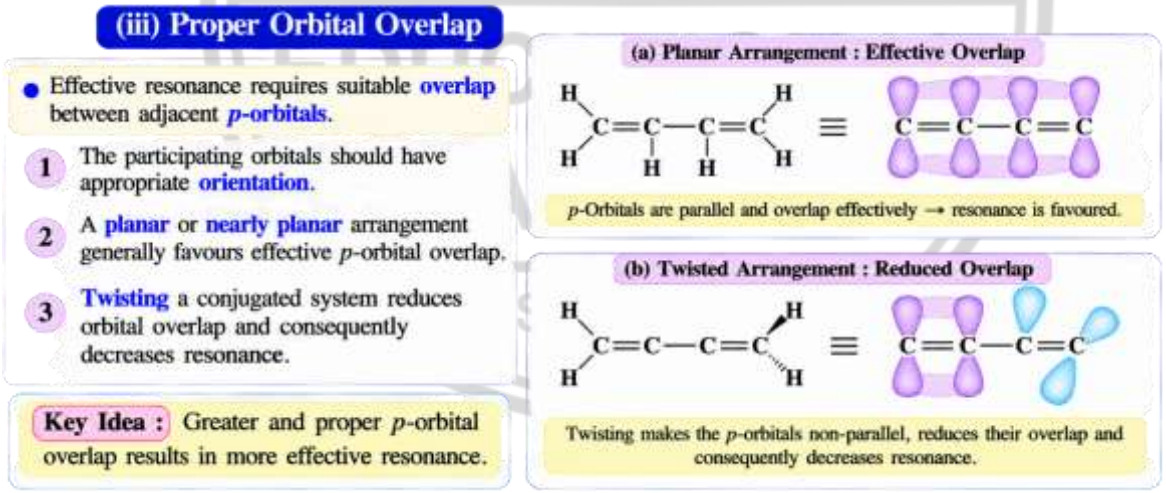
One of these lone pairs can overlap with the adjacent carbon p-orbital and participate in resonance.



(iii) Proper Orbital Overlap

Effective resonance requires suitable overlap between adjacent p-orbitals.

1. The participating orbitals should have appropriate orientation.
2. A planar or nearly planar arrangement generally favours effective p-orbital overlap.
3. Twisting a conjugated system reduces orbital overlap and consequently decreases resonance.



4.4 Types of Mesomeric Effect

Depending upon whether a group donates or withdraws electrons through resonance, the mesomeric effect is classified into two types:

1. Positive mesomeric effect (+M effect).
2. Negative mesomeric effect (−M effect).

4.5 Positive Mesomeric Effect (+M or +R Effect)

4.5.1 Definition

The permanent electron-displacement effect in which an atom or group donates electron density to an adjacent conjugated system through resonance is called the positive mesomeric effect (+M effect).

Groups showing the +M effect generally possess a lone pair of electrons that can participate in conjugation.

Important Points:

1. The group donates electron density through resonance.
2. The donating atom generally possesses at least one available lone pair.
3. Electron donation takes place through overlap of suitable orbitals.
4. The +M effect can increase electron density at particular positions of a conjugated system.
5. In aromatic compounds, common +M groups such as $-OH$ and $-NH_2$ increase electron density particularly at the ortho and para positions.
6. The +M effect can influence acidity, basicity, stability of intermediates, and electrophilic substitution reactions.

4.5.2 Common Groups Showing +M Effect

Group	Name
$-O^-$	Oxide/oxido substituent
$\mathbf{-NH_2}$	Amino
$\mathbf{-NHR}$	Alkylamino
$\mathbf{-NR_2}$	Dialkylamino
$\mathbf{-OH}$	Hydroxy
$\mathbf{-OR}$	Alkoxy
$\mathbf{-NHCOR}$	Acylamino
$\mathbf{-OCOR}$	Acyloxy

Group	Name
$-F$	Fluoro
$-Cl$	Chloro
$-Br$	Bromo
$-I$	Iodo

All these groups can donate electron density through resonance when attached to a suitable conjugated system, although their donating abilities differ considerably.

4.5.3 Explanation of +M Effect with an Example

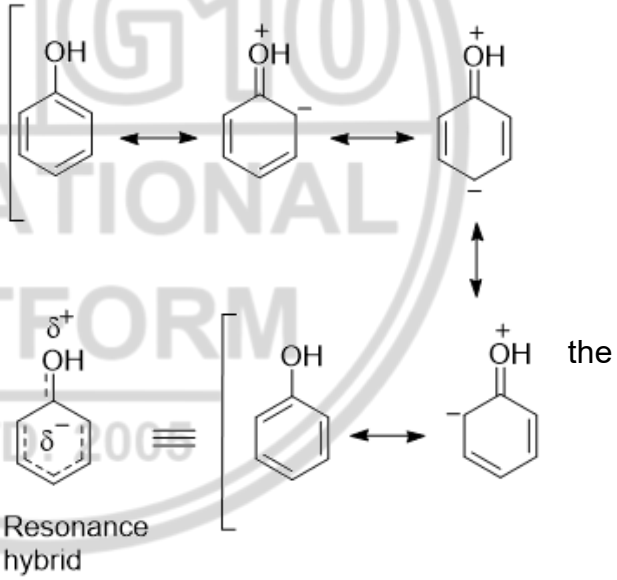
Example: Phenol (C_6H_5OH)

In phenol, the oxygen atom of the $-OH$ group possesses two lone pairs of electrons.

One lone pair can overlap with the π-orbitals of the benzene ring.

As a result, electron density is donated from oxygen into the aromatic ring through resonance.

C_6H_5OH



Explanation:

1. Oxygen donates a lone pair of electrons into the benzene ring.
2. This electron donation produces resonance contributors in which oxygen carries a positive formal charge.
3. Negative formal charge appears at ortho and para carbon atoms in the corresponding charge-separated contributors.
4. Consequently, the ortho and para positions acquire increased electron density in the resonance hybrid.
5. Therefore, the $-OH$ group generally activates the benzene ring towards electrophilic aromatic substitution and directs incoming electrophiles predominantly to the ortho and para positions.

Important Concept:

- Although oxygen is highly electronegative and exhibits a −I effect, it also shows a strong +M effect when attached directly to a benzene ring.
- For phenol, the electron-donating resonance effect dominates its influence on electrophilic aromatic substitution.
- Therefore, phenol is more reactive than benzene towards electrophilic aromatic substitution.

4.5.4 Order of +M Effect

- For common oxygen-containing substituents, the following qualitative comparison is useful:

$$-O^- > -OH, -OR$$

- For some commonly encountered electron-donating groups, a useful approximate order is:

$$\boxed{-O^- > -NH_2, \ -NHR, \ -NR_2 > -OH, \ -OR > -X}$$

Here, Xrepresents a halogen (F, Cl, Br or I).

Explanation:

1. A negatively charged oxygen atom in $-O^-$ has high electron density and can act as a very strong resonance donor.

2. The $-NH_2$ group donates its nitrogen lone pair into an adjacent conjugated system.

3. The $-OH$ and $-OR$ groups donate lone pairs from oxygen, but their donating strength is influenced by oxygen's high electronegativity.

4. The electron-donating ability of $-NHR$ and $-NR_2$ depends on inductive effects, molecular geometry, and steric interactions.

5. Halogens exhibit +M effects due to their lone pairs, but their strong −I effects generally make them deactivating groups in electrophilic aromatic substitution.

Important: There is no single universally valid +M strength order covering all substituents in every molecule. Their relative effects depend on the structure and the property being compared.

4.6 Negative Mesomeric Effect (−M or −R Effect)

4.6.1 Definition

- The permanent electron-displacement effect in which an atom or group withdraws electron density from an adjacent conjugated system through resonance is called the negative mesomeric effect (−M effect).

- Groups exhibiting the −M effect generally contain an electron-accepting π-system or a suitable vacant orbital capable of accepting electron density from the conjugated system.

Important Points:

1. The group withdraws electron density through resonance.
2. Such groups commonly contain multiple bonds involving electronegative atoms.
3. The −M effect decreases electron density at particular positions of the conjugated system.
4. In benzene derivatives, common −M groups decrease electron density particularly at the ortho and para positions.
5. Most common −M groups deactivate benzene rings towards electrophilic aromatic substitution.
6. The −M effect can stabilize certain negatively charged intermediates through electron delocalization.

4.6.2 Common Groups Showing −M Effect

Group	Name
$-NO_2$	Nitro
$-CN$	Cyano
$-CHO$	Formyl
$-COR$	Acyl
$-COOH$	Carboxyl
$-COOR$	Ester group
$-CONH_2$	Carboxamide group
$-SO_3H$	Sulfonic acid group

These groups generally withdraw electron density through resonance when directly attached to a suitable conjugated system.

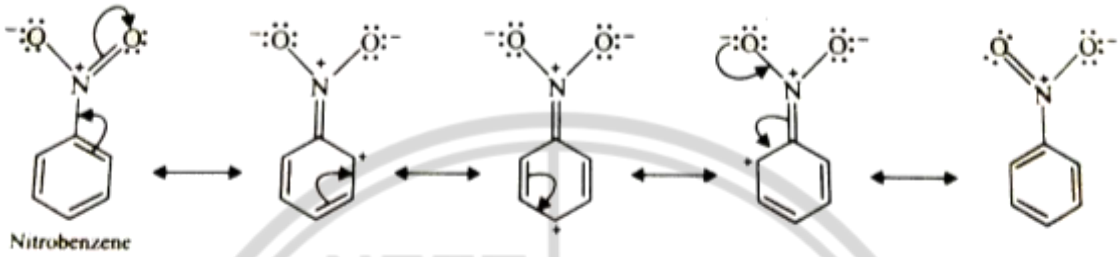
4.6.3 Explanation of −M Effect with an Example

Example: Nitrobenzene $(C_6H_5NO_2)$

The nitro group $(-NO_2)$ is a strong electron-withdrawing group.

When attached to a benzene ring, it withdraws electron density through both the −I effect and the −M effect.

$$C_6H_5NO_2$$



Explanation:

1. The nitro group contains a positively charged nitrogen atom in its usual formal-charge representation.
2. Its electron-accepting π-system can interact with the π-electrons of the benzene ring.
3. Electron density is delocalized from the benzene ring towards the nitro group.
4. The resonance contributors show positive formal charge at the ortho and para carbon atoms.
5. Consequently, electron density at these positions is reduced.
6. The benzene ring becomes less reactive towards electrophilic aromatic substitution.

Important Concept:

The $-NO_2$ group is a strongly deactivating, meta-directing group.

In electrophilic aromatic substitution, attack at the ortho or para position produces an arenium-ion intermediate with an especially unfavourable resonance contributor having positive charge on the carbon bonded to $-NO_2$.

Meta attack avoids this particular destabilizing contributor.

Therefore, nitrobenzene predominantly undergoes meta substitution.

4.6.4 Order of −M Effect

The strength of the −M effect depends upon the electron-accepting ability of the substituent and its interaction with the conjugated system.

For several common substituents, a useful qualitative trend is:

$$-NO_2, -CN: strong\ -M\ groups$$

$$-CHO, -COR, -COOR, -COOH: carbonyl\ based\ -M\ groups$$

Explanation:

1. The nitro group is a strong resonance electron acceptor.

2. The cyano group withdraws electron density through its polarized carbon–nitrogen triple bond.

3. Carbonyl-containing groups withdraw electron density through their polarized $C = O$ bonds.

4. The strength of withdrawal depends on additional substituents attached to the carbonyl carbon and on the molecular environment.

A strict universal ranking of all −M groups should not be assumed without specifying the system and the method of comparison.

4.7 Comparison Between +M and −M Effects

Basis	+M Effect	−M Effect
Nature	Electron donating	Electron withdrawing
Electron displacement	From substituent towards conjugated system	From conjugated system towards substituent
Typical requirement	Available lone pair for donation	Suitable electron-accepting orbital or π-system
Electron density	Increases at certain positions	Decreases at certain positions
Common examples	$-OH$, $-OR$, $-NH_2$	$-NO_2$, $-CN$, $-CHO$
Effect on benzene reactivity	Usually activating	Usually deactivating
Orientation in electrophilic aromatic substitution	Usually ortho/para directing	Usually meta directing

Exception: Halogens show a +M effect and direct electrophilic aromatic substitution to ortho and para positions, but they deactivate the benzene ring overall because of their strong −I effect.

5 Applications of resonance

5.1 Combined Effect of Inductive and Mesomeric Effects

In many organic compounds, both the inductive effect (I-effect) and the mesomeric effect (M-effect) operate simultaneously.

These two effects influence the distribution of electron density within a molecule and may act either in the same direction or in opposite directions.

Important Points:

1. The inductive effect involves the polarization of σ-bonds due to electronegativity differences between bonded atoms.

2. The mesomeric effect involves the delocalization of π-electrons or lone pairs through a suitable conjugated system.

3. Both effects are permanent electronic effects.

4. When both effects operate in the same direction, they reinforce each other.

5. When they operate in opposite directions, their relative strengths determine the overall electronic influence of the substituent.

6. In many conjugated systems, resonance has a stronger influence on particular properties than the inductive effect. However, this is not a universal rule.

5.1.1 When Inductive and Mesomeric Effects Act in the Same Direction

Certain groups withdraw electron density through both the −I and −M effects.

Examples include:

$$-NO_2, -CN, -CHO, -COOH, -COOR$$

Example: Nitrobenzene

The nitro group $(-NO_2)$ exhibits:

• −I effect: It withdraws electron density through σ-bonds.

• −M effect: It withdraws π-electron density from the benzene ring through resonance.

$$C_6H_5NO_2$$

Nitrobenzene: resonance illustration

Explanation:

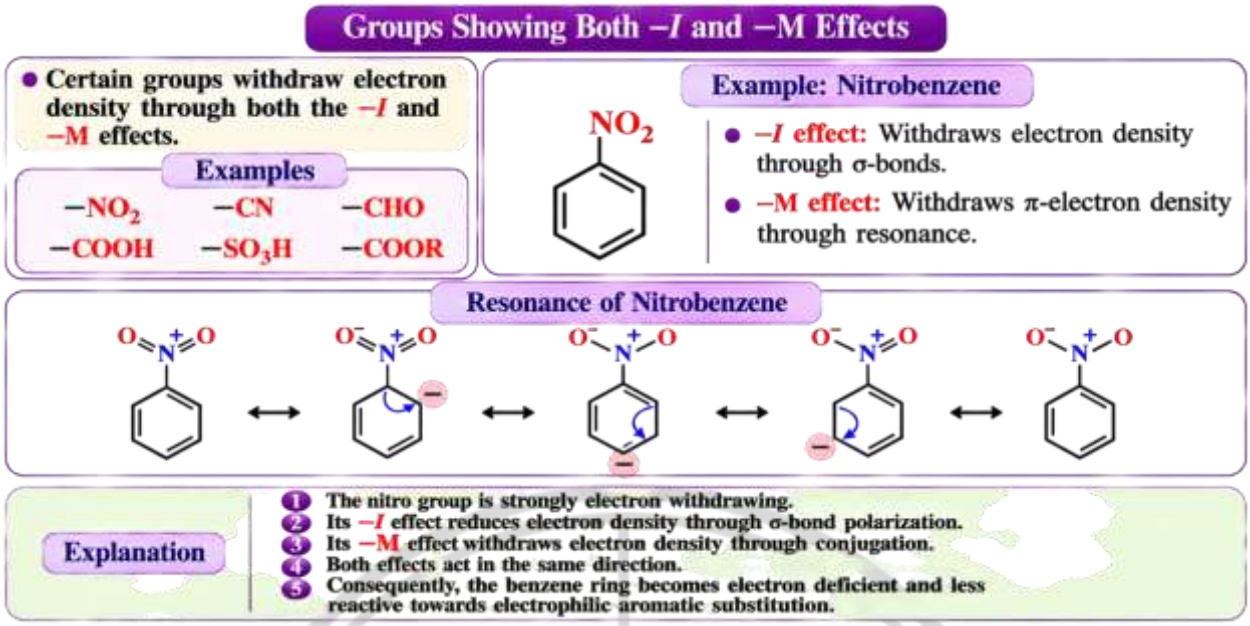
1. The nitro group is strongly electron withdrawing.

2. Its −I effect reduces electron density through σ-bond polarization.

3. Its −M effect withdraws electron density through conjugation with the benzene ring.

4. Both effects act in the same direction.

5. Consequently, the benzene ring becomes electron deficient and less reactive towards electrophilic aromatic substitution.



Important Concept: Nitrobenzene undergoes electrophilic aromatic substitution more slowly than benzene and predominantly gives meta-substituted products.

5.1.2 When Inductive and Mesomeric Effects Act in Opposite Directions

Some groups exhibit an electron-withdrawing inductive effect and an electron-donating mesomeric effect. Common examples are:

$$-OH, -OR, -NH_2, -F, -Cl, -Br, -I$$

These groups generally show:

$$-I\ effect\ and\ +M\ effect$$

However, the relative importance of these effects varies from one group to another.

Example 1: Phenol

Example 2: Chlorobenzene

Note: Both phenol and chlorobenzene exhibit the same types of electronic effects, namely the **negative inductive effect (−I)** and the **positive mesomeric effect (+M)**. However, their reactivities towards electrophilic aromatic substitution are opposite when compared with benzene.

- **In phenol:** The oxygen atom of the −OH group donates its lone pair into the benzene ring through effective overlap between oxygen's 2p orbital and carbon's 2p orbitals. The strong +M effect outweighs the deactivating −I effect. Consequently, **phenol is more reactive than benzene**.
- **In chlorobenzene:** Chlorine also donates a lone pair through resonance. However, its relatively diffuse 3p orbitals overlap less effectively with the carbon

2p orbitals of the benzene ring. Consequently, its resonance donation is comparatively weak and cannot overcome its overall electron-withdrawing influence. Therefore, **chlorobenzene is less reactive than benzene**.

However, chlorine remains an ortho-para directing group because its resonance donation stabilizes the ortho and para substitution intermediates relative to the meta intermediate.

5.2 Mesomeric and Inductive Effects in Vinyl Chloride

Vinyl chloride (CH_2=CH–Cl) is an important example of a compound in which chlorine exhibits both the negative inductive effect (−I) and the positive mesomeric effect (+M).

In vinyl chloride, chlorine is directly attached to an sp^2-hybridised carbon atom of the C=C double bond. Consequently, chlorine withdraws electron density through the σ-bond and donates electron density through resonance.

5.2.1 Inductive Effect (−I) in Vinyl Chloride

Chlorine is more electronegative than carbon. Therefore, it attracts the shared electrons of the C–Cl σ-bond towards itself, producing polarisation of the bond.

$$CH_2 = CH^{\delta+} - Cl^{\delta-}$$

Explanation:

1. Chlorine withdraws electron density through the C–Cl σ-bond.
2. The carbon attached to chlorine acquires a partial positive charge (δ^+).
3. Chlorine acquires a partial negative charge (δ^-) due to σ-bond polarisation.
4. This electron-withdrawing influence is known as the negative inductive effect (−I).

5.2.2 Mesomeric Effect (+M) in Vinyl Chloride

Chlorine possesses three lone pairs of electrons. One suitably oriented lone pair can overlap with the adjacent carbon–carbon π-system and participate in resonance.

The resonance structures can be represented as:

$$CH_2 = CH - Cl \leftrightarrow {}^{-}CH_2 - CH = Cl^{+}$$

Explanation:

1. Chlorine donates one of its lone pairs towards the adjacent carbon atom.
2. A π-bond contribution between carbon and chlorine appears in the charge-separated resonance structure.

3. Simultaneously, the original C=C π-electron pair shifts towards the terminal carbon atom.
4. Consequently, the terminal carbon carries a negative formal charge in this resonance contributor.
5. Chlorine carries a positive formal charge in the same contributor.
6. This electron donation through resonance is known as the positive mesomeric effect (+M) of chlorine.

5.2.3 Consequences of Resonance in Vinyl Chloride

1. Partial double-bond character: The C–Cl bond acquires some double-bond character due to resonance.

2. Bond length and strength: The C–Cl bond is shorter and generally stronger than an ordinary comparable C(sp^3)–Cl single bond.

3. Restricted rotation: Rotation about the C–Cl bond becomes more restricted because of its partial π-bond character.

4. Conjugation: A lone pair of chlorine participates in conjugation with the adjacent C=C double bond.

5. Chemical reactivity: Vinyl chloride is relatively resistant to ordinary nucleophilic substitution compared with typical alkyl chlorides.

Important clarification: The resonance contributor showing a positive formal charge on chlorine does not mean that chlorine necessarily carries a net positive partial charge in the actual molecule. Formal charges in individual resonance structures must be distinguished from the actual electron-density distribution in the resonance hybrid.

5.5 Difference Between Resonance and Mesomeric Effect

Resonance and mesomerism mean essentially the same thing: electron delocalisation. However, resonance is a general concept, whereas the mesomeric effect specifically describes the electron-donating or electron-withdrawing influence of an attached group through resonance.

5.5.1 Side-by-Side Comparison

Basis	Resonance	Mesomeric Effect (+M / −M)
1. Definition	Delocalisation of electrons within a molecule or ion.	Donation or withdrawal of electron density by a substituent through resonance.

2. Main purpose	Explains the actual electronic structure of a molecule.	Explains how an attached group influences electron distribution in a conjugated system.
3. Requirement	Requires suitable orbital overlap for electron delocalisation.	Requires a substituent conjugated with the rest of the molecule.
4. Electron movement	Electrons are delocalised over several atoms.	Electrons are donated or withdrawn by a particular group through conjugation.
5. Types	Not generally classified as +M or −M.	Two types: positive (+M) and negative (−M).
6. Substituent necessary?	No.	Yes, when discussing the electronic influence of a substituent.
7. Effect on stability	Delocalisation generally contributes to stability.	Can stabilise or destabilise particular intermediates depending on the system.
8. Examples	Benzene, 1,3-butadiene, carbonate ion.	−OH (+M), $-NH_2$ (+M), $-NO_2$ (−M), −CHO (−M).

5.5.2 Difference Through Examples (Side by Side)

Resonance	**Mesomeric Effect**
Example 1: Benzene-C_6H_6	Example 1: Phenol-C_6H_5OH
Six π-electrons are delocalised throughout the benzene ring.	Oxygen donates a lone pair into the benzene ring through resonance.
No substituent is needed for this delocalisation.	The −OH substituent is responsible for the +M effect.
Result: Equal C–C bond lengths and aromatic stabilisation.	Result: Increased electron density at ortho and para positions and activation towards electrophilic substitution.
Example 2: 1,3-Butadiene-$CH_2 = CH - CH = CH_2$	Example 2: Nitrobenzene-$C_6H_5NO_2$
π-electrons are delocalised over four carbon atoms.	The $-NO_2$ group withdraws electron density from the benzene ring through resonance.
There is no separate substituent whose +M or −M influence is being classified.	The $-NO_2$ substituent is responsible for the −M effect.

Result: The central C–C bond acquires partial double-bond character.	Result: The ring is deactivated towards electrophilic substitution.
Example 3: Carbonate ion-CO_3^{2-}	Example 3: Aniline-$C_6H_5NH_2$
The negative charge is delocalised over three oxygen atoms.	The $-NH_2$ group donates its lone pair into the benzene ring.
The delocalisation is described using three equivalent resonance contributors.	This electron donation is called the +M effect of $-NH_2$.
Result: Three equivalent C–O bonds and resonance stabilisation.	Result: Activation of the ring towards electrophilic substitution.

5.5.3 The Most Important Difference Using the Same Compound

Consider phenol (C_6H_5OH).

Resonance in phenol	**Mesomeric effect in phenol**
Describes the delocalisation of electrons over oxygen and the benzene ring.	Describes the electron-donating influence of the −OH group on the benzene ring.
Explains why several resonance contributors can be drawn.	Explains why −OH is classified as a +M group.
Focuses on the electronic structure of the entire molecule.	Focuses on the influence of the attached −OH substituent.
Question answered: How are electrons delocalised?	Question answered: Which group donates or withdraws electron density through resonance?

6 Hyperconjugation

6.1 Introduction

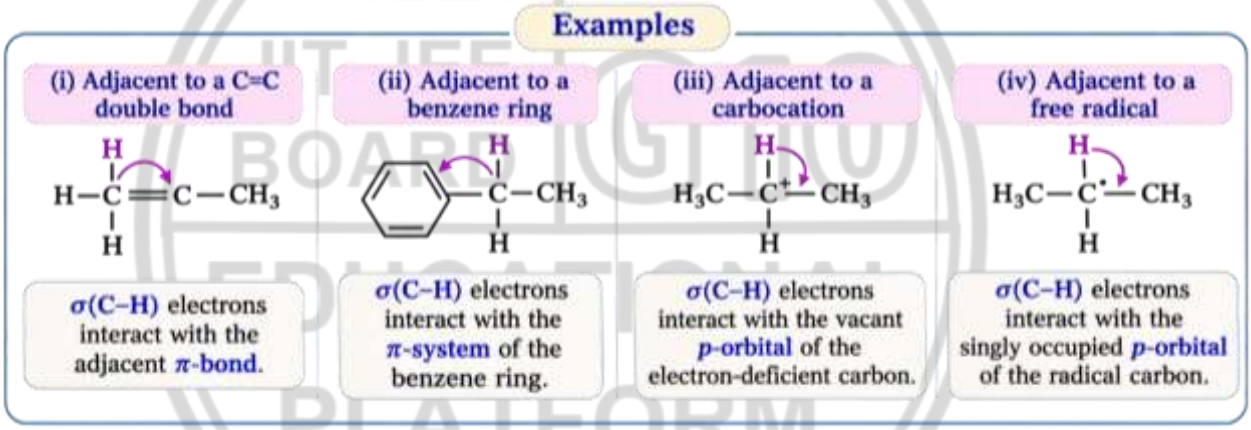
In organic compounds, electrons may become delocalized not only through π-bonds and lone pairs but also through certain σ-bonds.

- When a carbon–hydrogen $(C-H)$ σ-bond is adjacent to an unsaturated system, such as a carbon–carbon double bond, benzene ring, or an electron-deficient carbon atom (carbocation), the electrons of the σ-bond may interact with the neighbouring p-orbital or π-system.

- This interaction results in electron delocalization and is known as hyperconjugation. The concept is traditionally associated with Baker and Nathan and is also **called the Baker–Nathan effect.**

Important Points:

- Hyperconjugation involves the interaction of electrons of a σ-bond with an adjacent π-system or suitable p-orbital.
- In elementary organic chemistry, the donating σ-bond is usually a $C-H$ bond.
- Hyperconjugation is a permanent electronic effect.
- It can operate in alkenes, alkyl-substituted benzene rings, carbocations, and free radicals.



- It helps explain the stability, electron distribution, and chemical behaviour of several organic compounds.
- Hyperconjugation is **sometimes called no-bond resonance**, because some conventional resonance contributors are drawn without a bond between carbon and hydrogen.

6.2 Structural Requirements for Hyperconjugation

(i) Presence of an Adjacent π-System or Suitable p-Orbital

The molecule should contain a suitable electron-accepting orbital, such as:

- A π-system of an alkene.
- A π-system of an aromatic ring.
- A vacant p-orbital of a carbocation.
- A partially occupied p-orbital of a free radical.

For example:

$$CH_3 - CH = CH_2 \quad CH_3 - CH_2^+ \quad CH_3 - CH_2 \bullet$$

(ii) Presence of an α-$C-H$ Bond

In conventional hyperconjugation, the carbon atom directly adjacent to the unsaturated or electron-deficient centre should possess at least one hydrogen atom. This carbon is called the α-carbon, and its hydrogen atoms are called α-hydrogens.

$$CH_3 - CH = CH_2$$

The methyl carbon is the α-carbon with respect to the double bond, and it contains three α-hydrogens. Therefore, propene can exhibit $C-H$ hyperconjugation.

(iii) Proper Orbital Orientation

- Effective orbital overlap is necessary.
- Greater orbital overlap generally produces stronger hyperconjugative interaction.
- Molecular geometry and bond rotation can influence the extent of hyperconjugation.

6.5 Types of Hyperconjugation

Hyperconjugation can be classified according to the orbitals participating in electron delocalization.

6.5.1 σ–π Hyperconjugation

This occurs when electrons of a σ-bond interact with an adjacent π-system. Example: Propene

$$CH_3 - CH = CH_2$$

The $C-H$ σ-bonds of the methyl group interact with the adjacent carbon–carbon π-bond. This interaction contributes to the stability of the alkene.

6.5.2 σ–p Hyperconjugation

This occurs when a σ-bond interacts with an adjacent vacant or partially occupied p-orbital.

Example: Ethyl Carbocation

$$CH_3 - CH_2^+$$

The positively charged carbon possesses a vacant p-orbital. The adjacent methyl $C-H$ σ-bonds can donate electron density into this vacant orbital. This stabilizes the carbocation.

Example: Ethyl Free Radical

$$CH_3 - CH_2 \bullet$$

The radical carbon possesses a partially occupied p-orbital. Interaction with adjacent $C-H$ σ-bonds contributes to radical stabilization.

6.5.3 Negative Hyperconjugation (Reverse Hyperconjugation)

Negative hyperconjugation involves the donation of electron density from a filled orbital, such as a lone pair (n) or π-orbital, into an adjacent antibonding σ* orbital.

What Happens in Negative Hyperconjugation?

Consider a molecule containing:

1. A filled orbital, such as a lone pair on oxygen or nitrogen.
2. A nearby σ-bond, such as C–F or C–Cl.
3. An unoccupied antibonding σ* orbital associated with that bond.

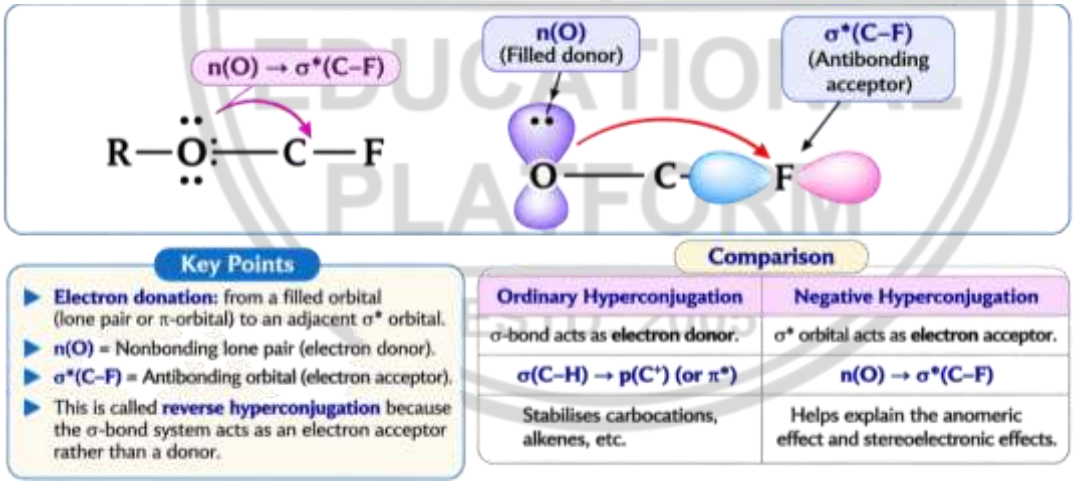
Every σ-bond has a corresponding antibonding orbital, represented by **σ***.

For example, the C–F bond has:

- **σ(C–F):** A filled bonding orbital.
- *σ(C–F):** An unoccupied antibonding orbital in the usual ground-state description.

If a neighbouring lone pair is suitably oriented, it can donate electron density into the σ*(C–F) orbital.

Negative Hyperconjugation:



Here, n(O) represents a nonbonding electron pair on oxygen.

- Thus, negative hyperconjugation involves electron donation **from a filled lone-pair orbital into a neighbouring σ-antibonding orbital**.

Why Is It Called Reverse Hyperconjugation?

In ordinary hyperconjugation, a filled σ-bond donates electron density into an adjacent vacant or partially occupied orbital. In negative hyperconjugation, the σ-bond system acts as an electron acceptor through its antibonding σ* orbital.

Comparison:

Ordinary Hyperconjugation	Negative Hyperconjugation

σ-bond acts as an electron donor.	σ* orbital acts as an electron acceptor.
Electrons are donated from a filled σ-orbital.	Electron density is donated into an antibonding σ* orbital.
Example: σ(C–H) → p(C^+)	Example: n(O) → σ*(C–F)
Helps stabilise carbocations and substituted alkenes.	Helps explain the anomeric effect and certain stereoelectronic effects.

Consequences of Negative Hyperconjugation

1. Electron donation into a σ* orbital can weaken and lengthen the corresponding σ-bond.
2. Favourable donor–acceptor orbital interaction can stabilise the overall molecule.
3. The strength of the interaction depends on effective orbital overlap and the relative orientation of the orbitals.

6.6 Number of Hyperconjugative Structures

- In the traditional no-bond resonance model, each suitable α-hydrogen may give rise to a separate hyperconjugative contributor.

$$Number\ of\ no-bond\ contributors\ =\ Number\ of\ \alpha-hydrogens$$

- Including the original conventional structure:

$$Total\ structures\ =\ Number\ of\ \alpha-hydrogens\ +\ 1$$

Example 1: Propene

$$CH_3-CH=CH_2$$

- Number of α-hydrogens = 3.

$$Total\ structures\ =\ 3\ +\ 1\ =\ 4$$

Example 2: 2-Methylpropene

$$CH_2=C(CH_3)_2$$

There are two methyl groups attached to the double-bonded carbon. Number of α-hydrogens:

$$3\ +\ 3\ =\ 6$$

$$Total\ structures\ =\ 6\ +\ 1\ =\ 7$$

6.7 Applications of Hyperconjugation

6.7.1 Stability of Alkenes

Hyperconjugation helps explain why alkyl-substituted alkenes are generally more stable than less substituted alkenes. Alkyl groups attached to a double bond can donate electron density through hyperconjugation.

Important Points:

- Alkyl groups can stabilize adjacent double bonds through hyperconjugation.
- A greater number of suitable α-C–H bonds often allows more hyperconjugative interactions.
- Alkyl substitution also affects alkene stability through other electronic and steric effects.
- Therefore, alkene stability cannot always be predicted using only the number of α-hydrogens.

General Stability Order: For many simple, acyclic alkenes:

$$Tetrasubstituted > Trisubstituted > Disubstituted > Monosubstituted > Ethene$$

This is a useful general trend rather than an absolute rule.

Example: Comparison of Monosubstituted Alkenes

(i) Propene

$$CH_3 - CH = CH_2$$

Number of α-hydrogens = 3.

(ii) But-1-ene

$$CH_3 - CH_2 - CH = CH_2$$

Number of α-hydrogens = 2.

(iii) 3-Methylbut-1-ene

$$(CH_3)_2CH - CH = CH_2$$

Number of α-hydrogens = 1.

- According to the simple α-hydrogen counting model, the hyperconjugative contribution is predicted to decrease as:

$$CH_3CH = CH_2 > CH_3CH_2CH = CH_2 > (CH_3)_2CHCH = CH_2$$

- This order compares the expected conventional $C - H$ hyperconjugative contribution, not necessarily the experimentally measured overall thermodynamic stability of the three alkenes.

Important Concept:

- For predicting alkene stability, consider degree of alkyl substitution, hyperconjugation, steric repulsion, geometrical isomerism, conjugation with other

π-systems, and ring strain where applicable. Trans-but-2-ene is generally more stable than cis-but-2-ene because of reduced steric repulsion.

- Therefore, we cannot reliably conclude that propene is the most stable and 3-methylbut-1-ene is the least stable merely by counting α-hydrogens.

6.7.2 Carbon–Carbon Double Bond Length in Alkenes

Hyperconjugation can influence bond lengths by changing the distribution of electron density and bond order. In an alkene, the $C = C$ bond contains one σ-bond and one π-bond.

Explanation:

- An adjacent C–H σ-bond interacts with the alkene π-system.
- Electron delocalization changes the localized π-bond character.
- The effective bond order and bond length may consequently change.
- Other structural factors, including substitution and steric effects, also influence the observed bond length.

Important: It is an oversimplification to state that increasing the number of hyperconjugative structures must always increase the $C = C$ bond length. Actual bond lengths depend on the complete molecular structure.

6.7.3 Stability of Alkyl Carbocations

Hyperconjugation is an important factor responsible for the stability of alkyl carbocations. A carbocation contains a positively charged carbon atom having an empty p-orbital. Adjacent $C - H$ σ-bonds can donate electron density into this empty orbital.

General Stability Order:

$$3° > 2° > 1° > CH_3^+$$

Explanation:

- Tertiary carbocations generally have more adjacent alkyl groups.
- Their neighbouring σ-bonds can participate in hyperconjugation.
- Alkyl groups also exert electron-donating inductive effects.
- These effects reduce electron deficiency at the positively charged carbon.
- Therefore, tertiary carbocations are generally more stable than secondary and primary carbocations.

Example: Number of α-Hydrogens

Carbocation	α-Hydrogens
$(CH_3)_3C^+$	9
$(CH_3)_2CH^+$	6

Carbocation	α-Hydrogens
$CH_3CH_2^+$	3
CH_3^+	0

Important Concept: Carbocation stability is not determined only by hyperconjugation. Resonance, inductive effects, aromaticity, and other structural factors can substantially change the stability order. Allylic and benzylic carbocations are stabilized by π-electron resonance.

6.7.4 Stability of Alkyl Free Radicals

A free radical contains an unpaired electron. In many carbon-centred radicals, the unpaired electron occupies a p-orbital or an orbital with substantial p-character. Adjacent σ-bonds can interact with this orbital through hyperconjugation.

General Stability Order:

$$3° > 2° > 1° > CH_3\bullet$$

Explanation:

- Alkyl groups provide hyperconjugative stabilization to adjacent radical centres.
- The unpaired electron density can become partially delocalized.
- This lowers the energy of the radical.
- Therefore, tertiary alkyl radicals are generally more stable than secondary and primary alkyl radicals.

Example:

$$(CH_3)_3C\bullet > (CH_3)_2CH\bullet > CH_3CH_2\bullet > CH_3\bullet$$

This is the usual qualitative stability order for simple alkyl radicals.

6.7.5 Electron-Releasing Power of Alkyl Groups in Alkylbenzenes

Alkyl groups attached to benzene rings generally increase the electron density of the aromatic system. For example, toluene contains a methyl group directly attached to benzene.

$$C_6H_5 - CH_3$$

The methyl group donates electron density through the +I effect and hyperconjugation.

Explanation:

- The methyl group contains three $C - H$ bonds adjacent to the aromatic π-system.
- These bonds can interact with the benzene π-orbitals.
- Hyperconjugation contributes to electron donation into the ring.

- Electron density increases particularly at the ortho and para positions in the conventional resonance representation.
- Therefore, the methyl group activates the benzene ring towards electrophilic aromatic substitution.

Electron-Releasing Power of Alkyl Groups: Consider the following groups attached directly to a benzene ring:

$$-CH_3, -CH_2CH_3, -CH(CH_3)_2, -C(CH_3)_3$$

Alkyl group	α-Hydrogens
$-\mathbf{CH_3}$	3
$-\mathbf{CH_2CH_3}$	2
$-\mathbf{CH(CH_3)_2}$	1
$-\mathbf{C(CH_3)_3}$	0

According to the conventional $\mathrm{C-H}$ hyperconjugation model, their hyperconjugative donation tendency follows:

$$-CH_3 > -CH_2CH_3 > -CH(CH_3)_2 > -C(CH_3)_3$$

However, the electron-donating inductive effect generally increases with increasing alkyl substitution.

Important distinction: The above order describes the simple α-$\mathrm{C-H}$ hyperconjugation model, not a universal experimental order of electron-releasing ability or aromatic reactivity. Other σ-bond interactions, steric effects, and reaction conditions can alter the observed trends.

6.7.6 Heat of Hydrogenation

The heat of hydrogenation is the heat released when an unsaturated compound reacts with hydrogen to form a saturated compound. For example:

$$CH_2 = CH_2 + H_2 \rightarrow CH_3CH_3$$

Hydrogenation of a typical alkene is exothermic.

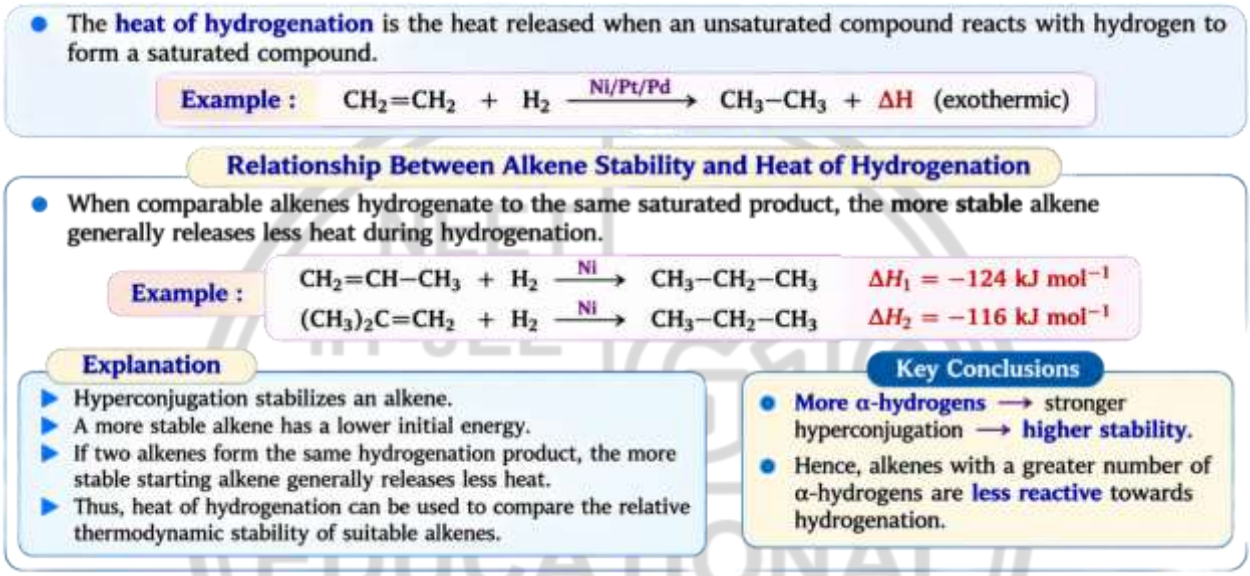
Relationship Between Alkene Stability and Heat of Hydrogenation: When comparable alkenes hydrogenate to the same saturated product, the **more stable alkene** generally releases **less heat during hydrogenation**.

$$Greater\ alkene\ stability \Rightarrow Smaller\ magnitude\ of\ heat\ of\ hydrogenation$$

Explanation:

- Hyperconjugation can stabilize an alkene.
- A more stable alkene has a lower initial energy.

- If two alkenes form the same hydrogenation product, the more stable starting alkene generally releases less heat, and show is **less reactive**
- Thus, **heat of hydrogenation can be used to compare the relative thermodynamic stability** of suitable alkenes.



Important Concept The stability of an alkene does not depend only on the number of α-hydrogens or hyperconjugation. Steric effects and geometrical arrangement also influence stability.

For example, cis-but-2-ene and trans-but-2-ene have the same number of α-hydrogens (6), but trans-but-2-ene is more stable because of reduced steric repulsion.

6.7.7 Effect of Hyperconjugation on Dipole Moment

Hyperconjugation can influence the distribution of electron density within a molecule and consequently affect its dipole moment.

Dipole moment is a measure of the separation of positive and negative charges.

$$\mu = q \times d$$

where μ = dipole moment; q = magnitude of separated charge; d = distance between the centres of charge.

Example: Formaldehyde and Acetaldehyde

Formaldehyde:

$$HCHO \quad \mu \approx 2.27\ D$$

Acetaldehyde:

$$CH_3CHO \quad \mu \approx 2.72\ D$$

The methyl group in acetaldehyde can interact with the carbonyl π-system through hyperconjugation.

Explanation:

- Formaldehyde contains a polar $C = O$ bond.
- Acetaldehyde also contains a polar $C = O$ bond.
- In acetaldehyde, the adjacent methyl group introduces additional electronic interactions.
- Hyperconjugation can contribute to changes in the electron distribution of the carbonyl group.
- Consequently, the overall molecular dipole moment differs from that of formaldehyde.

Important: The dipole moment of a molecule depends on its complete electron distribution and molecular geometry. The difference between the dipole moments of formaldehyde and acetaldehyde cannot be attributed exclusively to hyperconjugation.

6.7.8 Orienting Influence of Alkyl Groups in Benzene

Alkyl groups are generally activating and ortho-para directing in electrophilic aromatic substitution. Example: Toluene

$$C_6H_5CH_3$$

When toluene undergoes nitration, the major products are ortho-nitrotoluene and para-nitrotoluene.

Explanation:

- The methyl group releases electron density through its +I effect.
- It also participates in hyperconjugative interactions with the aromatic ring.
- These effects make the benzene ring more electron rich than benzene itself.
- Electrophilic attack at the ortho and para positions produces arenium-ion intermediates that are particularly stabilized by the alkyl substituent.
- Therefore, the methyl group directs incoming electrophiles mainly towards the ortho and para positions.

Example :

$$-CH_3: Activating, ortho - para\ directing$$

Orienting Influence of the $-CCl_3$ Group: The trichloromethyl group ($-CCl_3$) has a strong electron-withdrawing inductive effect. It generally deactivates the benzene ring and directs electrophilic substitution predominantly towards the meta position.

$$-CCl_3: Deactivating, meta\ directing$$

This behaviour is commonly explained primarily by its strong −I effect, with additional stereoelectronic interactions considered in more advanced treatments.

6.9 Comparison Between Hyperconjugation and Resonance

Basis	Hyperconjugation	Conventional π-Resonance
Electron involvement	Mainly σ-bond electrons	Mainly π-electrons or lone pairs
Orbital interaction	σ–π, σ–p, or related interactions	π–π or lone-pair–π interactions
Typical example	Propene	1,3-Butadiene
Structural requirement	Suitable σ-bond adjacent to an accepting orbital	Continuous conjugated orbital system
Conventional representation	No-bond resonance contributors	Canonical resonance structures
Main consequence	Electron delocalization and stabilization	Electron delocalization and stabilization

Both phenomena involve electron delocalization, but the participating orbitals and the conventional methods of representation differ.

7 Inductomeric and electromeric effects

7.1 Inductomeric Effect

7.1.1 Definition

The inductomeric effect is a temporary electronic effect in which the polarization of a σ-bond is increased due to the approach of an attacking reagent.

- It is considered a temporary enhancement of the inductive effect and operates only under the influence of an approaching reagent.

Important Points:

1. The inductomeric effect is a temporary electronic effect.
2. It involves increased polarization of electrons in a σ-bond.
3. It occurs due to the influence of an approaching attacking reagent.
4. It is associated with the inductive effect, but unlike the permanent inductive effect, the additional polarization is temporary.
5. The effect is significant in understanding the electronic changes that occur during certain organic reactions.

6. The additional polarization decreases when the external influence is removed, provided no chemical reaction has taken place.

7.1.2 Example: Reaction of Methyl Chloride with Hydroxide Ion

Consider methyl chloride:

$$\mathrm{CH_3 - Cl}$$

The chlorine atom is more electronegative than carbon. Therefore, chlorine attracts the shared pair of electrons towards itself.

The $\mathrm{C - Cl}$ bond becomes polarized:

$$CH_3{}^{\delta+} - Cl^{\delta-}$$

This polarization is due to the −I effect of chlorine.

When a hydroxide ion approaches methyl chloride, the electron distribution changes further as the substitution reaction proceeds.

The reaction is:

$$CH_3Cl + OH^- \rightarrow CH_3OH + Cl^-$$

Explanation:

1. In methyl chloride, chlorine withdraws electron density through the $\mathrm{C - Cl}$ σ-bond because of its greater electronegativity.

2. Consequently, the carbon atom acquires a partial positive charge.

3. The hydroxide ion, being negatively charged and electron rich, approaches the carbon atom.

4. As hydroxide approaches, the electronic distribution in the $\mathrm{C - Cl}$ bond changes.

5. A new $\mathrm{C - O}$ bond begins to form while the $\mathrm{C - Cl}$ bond begins to break.

6. The bonding electron pair of the $\mathrm{C - Cl}$ bond ultimately moves to chlorine.

7. Methanol is formed, and chloride ion leaves.

7.1.3 Relationship Between Inductive and Inductomeric Effects

Basis	Inductive Effect	Inductomeric Effect
Nature	Permanent	Temporary
Cause	Electronegativity differences and substituent effects	Influence of an approaching reagent
Electrons involved	σ-bond electrons	σ-bond electrons

Basis	Inductive Effect	Inductomeric Effect
Reagent requirement	Not required	Required
Main consequence	Permanent bond polarization	Additional polarization during reagent approach
Example	Polarization of $C-Cl$ bond in CH_3Cl	Additional polarization during attack by OH^-

Important Concept: The term inductomeric effect is mainly found in traditional organic chemistry textbooks. Modern reaction mechanisms generally describe the same situation using bond polarization, transition-state development, and electron redistribution during the reaction. In the methyl chloride example, the reaction proceeds by an SN2 mechanism, not by the formation of a free methyl carbocation.

7.2 Electromeric Effect

7.2.1 Definition

The electromeric effect (E-effect) is a temporary electronic effect involving the complete transfer of a π-electron pair of a multiple bond towards one of the bonded atoms under the influence of an attacking reagent.

- It is observed in compounds containing multiple bonds, particularly carbon–carbon double bonds and carbonyl groups.

Important Points:

1. The electromeric effect is a temporary electronic effect.
2. It occurs in compounds containing multiple bonds.
3. It involves the complete displacement of a π-electron pair in the conventional representation.
4. The effect is produced under the influence of an attacking reagent.
5. The electron-pair movement is represented by a curved arrow.
6. The effect may produce charge separation or facilitate the formation of a new bond.
7. It is classified into positive electromeric effect (+E) and negative electromeric effect $(-E)$.

7.2.2 Representation of Electromeric Effect

Consider a generalized multiple bond:

$$A = B$$

Under the influence of an attacking reagent, the π-electron pair may be transferred towards one atom. For example:

$$A = B \rightarrow A^{+} - B^{-}$$

Here, the π-electron pair is formally transferred towards B.

Explanation:

1. Initially, A and B are connected by a multiple bond.
2. The attacking reagent influences the electron distribution.
3. The π-electron pair shifts towards one of the bonded atoms.
4. One atom becomes electron deficient, while the other becomes electron rich in the formal representation.
5. The resulting electronic arrangement facilitates the subsequent reaction.

Important: The displayed charged structure is a formal representation of electron-pair movement. It does not necessarily represent an independently existing intermediate.

7.3 Types of Electromeric Effect

Depending on the direction of electron-pair movement relative to the attacking reagent, the electromeric effect is classified into:

1. Positive electromeric effect (+E effect).
2. Negative electromeric effect (−E effect).

7.3.1 Positive Electromeric Effect (+E Effect)

Definition

The temporary electronic effect in which the π-electron pair of a multiple bond is transferred towards the atom to which the attacking reagent becomes attached is called the positive electromeric effect (+E effect).

It is commonly illustrated by the addition of an electrophile to an alkene.

Example: Addition of H^{+} to Propene

Consider propene:

$$\mathrm{CH_3 - CH = CH_2}$$

Propene contains a carbon–carbon double bond. When a proton approaches the double bond, the π-electrons can participate in forming a new $\mathrm{C - H}$ bond.

The reaction may be represented as:

$$CH_3 - CH = CH_2 + H^{+} \rightarrow CH_3 - CH^{+} - CH_3$$

Explanation:

1. Propene contains a π-bond rich in electrons.

2. The proton (H^+) acts as an electrophile because it is electron deficient.

3. The π-electrons of the double bond are attracted towards the proton.

4. A new $C-H$ σ-bond forms between the proton and the terminal carbon atom.

5. The other double-bonded carbon develops a positive charge.

6. The major carbocation formed is the secondary carbocation, which is more stable than the alternative primary carbocation.

The carbocation is:

$$CH_3-CH^+-CH_3$$

Why Is the Secondary Carbocation Formed Preferentially?

Two carbocations are theoretically possible during protonation of propene.

(i) Primary carbocation:

$$CH_3-CH_2-CH_2{}^+$$

(ii) Secondary carbocation:

$$CH_3-CH^+-CH_3$$

The secondary carbocation is more stable because:

1. Two methyl groups donate electron density through the +I effect.

2. Adjacent $C-H$ σ-bonds provide hyperconjugative stabilization.

3. The positive charge is better stabilized than in the primary carbocation.

Therefore, protonation preferentially produces the secondary carbocation.

IIT-JEE Important Concept: This regioselectivity is consistent with Markovnikov's rule for the usual ionic addition of hydrogen halides to unsymmetrical alkenes. The π-electron pair is used to form the new bond with the electrophile. It should not be interpreted as first moving to an isolated carbon atom and then separately attaching to H^+.

7.3.2 Negative Electromeric Effect (−E Effect)

Definition

The temporary electronic effect in which the π-electron pair of a multiple bond is transferred away from the atom to which the attacking reagent becomes attached is called the negative electromeric effect (−E effect).

It is commonly illustrated by the nucleophilic addition of a reagent to a carbonyl compound.

Example: Addition of Cyanide Ion to a Carbonyl Compound

Consider a carbonyl compound:

$$R_2C=O$$

The carbonyl group is polarized because oxygen is more electronegative than carbon. Therefore:

$$C^{\delta+} = O^{\delta-}$$

When cyanide ion (CN^-) approaches the carbonyl carbon, it attacks the electron-deficient carbon atom.

The reaction is:

$$R_2C = O + CN^- \rightarrow R_2C(CN) - O^-$$

Explanation:

1. The carbonyl carbon carries a partial positive charge.
2. The cyanide ion is a nucleophile and possesses an electron pair available for bond formation.
3. The carbon end of cyanide attacks the carbonyl carbon.
4. Simultaneously, the π-electron pair of the $C = O$ bond shifts towards oxygen.
5. A new carbon–carbon σ-bond forms between the carbonyl carbon and cyanide.
6. Oxygen acquires a negative formal charge, forming an alkoxide intermediate.
7. Subsequent protonation of the alkoxide produces a cyanohydrin.

The protonation step is:

$$R_2C(CN) - O^- + H^+ \rightarrow R_2C(OH) - CN$$

Important Concept: Nucleophilic addition to aldehydes and ketones is a fundamental reaction of carbonyl compounds. The movement of π-electrons towards oxygen is a key step in the mechanism.

7.4 Comparison Between +E and −E Effects

Basis	+E Effect	−E Effect
Direction of π-electron movement	Towards the atom to which the reagent attaches	Away from the atom to which the reagent attaches
Common attacking reagent	Electrophile	Nucleophile
Typical substrate	Alkene	Carbonyl compound
Example	Protonation of propene	Addition of CN^- to a carbonyl compound
New bond formation	Electrophile bonds to a carbon of the double bond	Nucleophile bonds to carbonyl carbon

Basis	+E Effect	−E Effect
Electron-pair role	π-electrons form a bond with electrophile	π-electrons shift towards oxygen

7.5 Direction of Electron-Pair Shift

The direction of electron-pair movement depends on the nature of the multiple bond, the attached substituents, and the attacking reagent.

7.5.1 When Similar Groups Are Attached to the Multiple Bond

If the two ends of a multiple bond are chemically equivalent, electron-pair movement towards either end may lead to equivalent products.

Example: Ethene

$$\mathrm{CH_2 = CH_2}$$

The two carbon atoms are equivalent. During addition of $\mathrm{H^+}$, protonation at either carbon gives the same ethyl carbocation.

$$CH_2 = CH_2 + H^+ \rightarrow CH_3 - CH_2{}^+$$

Therefore, there is no regioselectivity between the two carbon atoms.

7.5.2 When Dissimilar Groups Are Attached to the Multiple Bond

If the groups attached to the two ends of a multiple bond are different, the direction of electron-pair movement may be influenced by:

1. Inductive effects of the substituents.
2. Resonance effects.
3. Hyperconjugation.
4. Stability of the intermediate or transition state.
5. Nature of the attacking reagent.
6. Reaction conditions.

Example: Propene

$$\mathrm{CH_3 - CH = CH_2}$$

The methyl group donates electron density through the +I effect and hyperconjugation. During ordinary electrophilic protonation, the pathway leading to the more stable secondary carbocation is preferred.

$$CH_3 - CH = CH_2 + H^+ \rightarrow CH_3 - CH^+ - CH_3$$

7.5.3 Direction of Electron-Pair Shift in Carbonyl Compounds

In a carbonyl group:

$$C = O$$

Oxygen is more electronegative than carbon. Therefore, the bond is polarized:

$$C^{\delta+} = O^{\delta-}$$

Carbonyl polarisation can also be illustrated by the following resonance contributors:

$C = O \leftrightarrow C^{+} - O^{-}$

These contributors describe one carbonyl group. During nucleophilic addition, a curved arrow from the $C = O$ bond to oxygen shows electron-pair movement as the nucleophile forms its bond to carbon.

Explanation:

1. Oxygen has a greater tendency to attract electron density than carbon.
2. The carbonyl carbon is electrophilic.
3. A nucleophile attacks the carbonyl carbon.
4. The π-electrons move towards oxygen to avoid exceeding the normal valency of carbon.
5. An alkoxide intermediate is formed.

Important: The statement that electrons always shift towards oxygen applies to the usual nucleophilic addition mechanism of carbonyl compounds. Other reaction mechanisms involving carbonyl groups may have different electron-flow patterns.

7.6 Relationship Between Inductive and Electromeric Effects

In compounds containing polar multiple bonds, inductive and electromeric effects may operate together. Consider the carbonyl group:

$$C = O$$

Oxygen withdraws electron density because of its greater electronegativity. This produces permanent polarization of the carbonyl bond. During nucleophilic attack, the π-electron pair shifts towards oxygen.

Explanation:

1. The inductive effect contributes to the permanent polarization of the carbonyl group.
2. The carbonyl carbon becomes electron deficient.
3. The nucleophile is attracted towards the carbonyl carbon.
4. During bond formation, the π-electron pair shifts towards oxygen.
5. This temporary electron movement is traditionally described as the −E effect.

In traditional textbook terminology, the electromeric effect is often described as predominating over the inductive effect during such reactions because it involves complete formal displacement of a π-electron pair. However, the two effects describe

different aspects of electron distribution and should not be treated as universally competing effects with a fixed strength order.

7.7 Difference Between Inductive, Inductomeric, and Electromeric Effects

Basis	Inductive Effect	Inductomeric Effect	Electromeric Effect
Nature	Permanent	Temporary	Temporary
Electrons involved	σ-electrons	σ-electrons	π-electrons
Cause	Bond polarity and substituent influence	Approach of an attacking reagent	Approach of an attacking reagent
Multiple bond necessary	No	No	Yes
Complete π-electron-pair transfer	No	No	Yes, in the conventional model
Example	$C-Cl$ bond polarization	Reagent-induced polarization of CH_3Cl	Addition to $C=C$ or $C=O$

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8 Bond fission and reactive intermediates

8.1 Introduction

- During organic chemical reactions, existing covalent bonds may break and new covalent bonds may form.
- A covalent bond is formed by the sharing of electrons between two atoms. When this bond breaks, the shared electron pair must be distributed between the resulting species.
- The process of breaking a covalent bond is known as bond fission or bond cleavage.
- Depending upon how the shared pair of electrons is distributed, covalent bond cleavage is classified into two types:

1. Homolytic bond fission (Homolysis)
2. Heterolytic bond fission (Heterolysis)

- The nature of bond cleavage determines the type of reactive intermediate formed and strongly influences the mechanism of an organic reaction.

8.2 Homolytic Bond Fission (Homolysis)

8.2.1 Definition

- Homolytic bond fission is the process in which a covalent bond breaks symmetrically so that each bonded atom takes one electron from the shared electron pair.
- As a result, two species containing unpaired electrons are formed.
- These species are called free radicals.

General Representation

$$A - B \rightarrow A\bullet + B\bullet \quad (h\nu \text{ or } \Delta)$$

- A−B represents a covalent bond.
- A• represents a radical containing an unpaired electron.
- B • represents another radical containing an unpaired electron.
- hv represents energy supplied by light.
- Δ represents heating.

$$\mathrm{A - B \rightarrow A\bullet + B\bullet}$$

Schematic representation of homolytic cleavage

8.2.2 Explanation of Homolytic Bond Fission

1. Initially, two atoms are connected by a covalent bond containing two shared electrons.

2. During homolysis, the bond breaks symmetrically.

3. Each atom receives one electron from the original bonding pair.

4. The resulting fragments contain unpaired electrons.

5. Such fragments are known as free radicals.

6. The reaction generally requires energy to break the covalent bond.

Important Concept: Homolysis produces radicals, but the resulting fragments are not necessarily electrically charged.

For example:

$$Cl_2 \rightarrow 2Cl\bullet$$

Both chlorine radicals are electrically neutral.

8.2.3 Example: Homolytic Cleavage of Chlorine Molecule

A chlorine molecule contains a covalent bond between two chlorine atoms.

When chlorine is exposed to suitable ultraviolet light, the $\mathrm{Cl}-\mathrm{Cl}$ bond undergoes homolytic cleavage.

$$Cl-Cl \rightarrow Cl\bullet + Cl\bullet \quad (h\nu)$$

$$\mathrm{Cl}-\mathrm{Cl} \rightarrow \mathrm{Cl}\bullet + \mathrm{Cl}\bullet$$

Chlorine homolysis under light

Explanation

1. The chlorine molecule absorbs light energy.

2. The $\mathrm{Cl}-\mathrm{Cl}$ covalent bond breaks.

3. Each chlorine atom receives one electron from the bonding pair.

4. Two chlorine radicals are produced.

5. These radicals are highly reactive and can initiate chain reactions.

8.2.4 Representation Using Fish-Hook Arrows

In organic reaction mechanisms, homolytic bond cleavage is represented by single-headed curved arrows, commonly called fish-hook arrows.

Each fish-hook arrow represents the movement of one electron.

Important Points:

1. Two fish-hook arrows are used to represent the homolytic cleavage of a single covalent bond.

2. One arrow shows the movement of one bonding electron towards the first atom.

3. The other arrow shows the movement of the second bonding electron towards the second atom.

4. This representation distinguishes homolytic cleavage from heterolytic cleavage, which involves movement of an electron pair.

8.3 Factors Favouring Homolytic Bond Fission

8.3.1 Small Electronegativity Difference

Homolytic cleavage is commonly favoured when the electronegativity difference between the bonded atoms is small or zero.

For example:

$$Cl - Cl, Br - Br, C - C$$

These bonds involve atoms of identical or similar electronegativity.

Therefore, the bonding electrons are shared relatively equally.

However, the electronegativity difference alone does not determine the mode of bond cleavage.

8.3.2 Non-Polar Solvents

Homolytic reactions are often favoured in non-polar solvents because such solvents generally do not strongly stabilize separated ions.

Examples include:

- Benzene
- Hexane
- Carbon tetrachloride

A non-polar medium may therefore favour radical pathways over pathways requiring highly separated ions, depending on the reaction.

8.3.3 Presence of Heat

Heat can provide the energy required for homolytic bond cleavage.

For example, suitable organic peroxides undergo thermal decomposition to form radicals.

$$RO - OR \rightarrow 2RO\bullet \quad (\Delta)$$

The resulting radicals can initiate further reactions.

8.3.4 Presence of Light

Light, particularly ultraviolet radiation, can supply energy for bond cleavage.

For example:

$$Cl_2 \rightarrow 2Cl\bullet \quad (h\nu)$$

This process is important in photochemical reactions and free-radical halogenation.

8.3.5 Presence of Peroxides

Peroxides contain a relatively weak oxygen–oxygen bond.

On heating or irradiation, this bond may undergo homolytic cleavage.

$$ROOR \rightarrow 2RO\bullet \quad (\Delta \text{ or } h\nu)$$

The radicals produced can initiate radical chain reactions.

8.3.6 Electrical Energy

Electrical energy may initiate bond cleavage by producing energetic electrons, excited species, or radicals.

For example, electrical discharge can initiate radical formation in suitable gases.

IIT-JEE Important Concept: Homolytic cleavage is particularly important in photochemical reactions, peroxide-initiated reactions, and free-radical chain reactions.

8.4 Free Radicals

8.4.1 Definition

A free radical is an atom or group of atoms containing one or more unpaired electrons and capable of independent existence, at least for a short time.

In introductory organic chemistry, free radicals generally refer to species containing one unpaired electron.

Examples:

$$Cl\bullet, CH_3\bullet, C_2H_5\bullet$$

8.4.2 Characteristics of Free Radicals

1. Free radicals contain unpaired electrons.
2. Most simple organic radicals are highly reactive.
3. They are often formed by homolytic bond cleavage.
4. They may be electrically neutral, although radical ions also exist.
5. Carbon-centred radicals are often approximately trigonal planar.
6. Their stability depends on resonance, hyperconjugation, inductive effects, and other structural factors.

8.4.3 Stability of Alkyl Free Radicals

For simple alkyl radicals, the general stability order is:

$$3^\circ > 2^\circ > 1^\circ > CH_3\bullet$$

Explanation:

1. Alkyl groups can stabilize an adjacent radical centre through hyperconjugation.

2. Greater alkyl substitution generally provides greater stabilization.

3. Consequently, tertiary alkyl radicals are usually more stable than secondary and primary alkyl radicals.

Important: Allylic and benzylic radicals are also stabilized by resonance.

8.5 Homolytic Reaction Mechanism (Free-Radical Mechanism)

A reaction mechanism involving radical intermediates is called a free-radical mechanism.

A typical free-radical chain reaction consists of three stages:

1. Initiation
2. Propagation
3. Termination

Example: Chlorination of Methane

The overall reaction is:

$$CH_4 + Cl_2 \rightarrow CH_3Cl + HCl \quad (h\nu)$$

(i) Initiation

Chlorine undergoes homolytic cleavage in the presence of light.

$$Cl_2 \rightarrow 2Cl\bullet \quad (h\nu)$$

(ii) Propagation

A chlorine radical abstracts a hydrogen atom from methane.

$$Cl\bullet + CH_4 \rightarrow HCl + CH_3\bullet$$

The methyl radical then reacts with chlorine.

$$CH_3\bullet + Cl_2 \rightarrow CH_3Cl + Cl\bullet$$

A new chlorine radical is formed, allowing the chain reaction to continue.

(iii) Termination

Radicals combine to form stable molecules.

Examples:

$$Cl\bullet + Cl\bullet \rightarrow Cl_2$$

$$CH_3\bullet + Cl\bullet \rightarrow CH_3Cl$$

$$CH_3\bullet + CH_3\bullet \rightarrow C_2H_6$$

IIT-JEE Important Concept: Free-radical chain reactions involve radical formation, radical propagation, and radical termination. Understanding these stages is essential for alkane halogenation and peroxide-initiated reactions.

8.6 Heterolytic Bond Fission (Heterolysis)

8.6.1 Definition

Heterolytic bond fission is the process in which a covalent bond breaks unsymmetrically so that one bonded atom receives both electrons of the shared electron pair.

As a result, oppositely charged species are formed.

These species are called ions.

General Representation

For a covalent bond A–B:

$$A - B \rightarrow A^+ + B^-$$

In this representation, both bonding electrons move towards B.

Alternatively:

$$A - B \rightarrow A^- + B^+$$

if the bonding electrons move towards A.

$$\mathrm{A - B \rightarrow A^+ + B^-}$$

Schematic representation of heterolytic cleavage

8.6.2 Explanation of Heterolytic Bond Fission

1. Initially, two atoms share a pair of electrons in a covalent bond.
2. During heterolysis, both bonding electrons move towards one atom.
3. The atom receiving both electrons becomes negatively charged, if the original species was neutral.
4. The atom losing its share of the bonding electrons becomes positively charged.
5. Therefore, heterolysis of a neutral covalent bond produces a cation and an anion.
6. The process is favoured when the resulting ions are sufficiently stabilized.

8.6.3 Example: Heterolytic Cleavage of a Carbon–Chlorine Bond

Consider methyl chloride:

$$\mathrm{CH_3 - Cl}$$

A formal heterolytic cleavage may be represented as:

$$CH_3 - Cl \rightarrow {CH_3}^+ + Cl^-$$

Explanation:

1. Chlorine is more electronegative than carbon.
2. The $\mathrm{C - Cl}$ bond is polarized towards chlorine.
3. During formal heterolysis, both bonding electrons move towards chlorine.

4. Chloride ion is formed.

5. The carbon-containing fragment becomes a carbocation.

Important clarification: The equation illustrates heterolytic bond cleavage. It does not imply that methyl chloride readily produces a free methyl carbocation under ordinary conditions. Methyl chloride generally undergoes nucleophilic substitution by an SN2 mechanism.

8.7 Factors Favouring Heterolytic Bond Fission

8.7.1 Large Electronegativity Difference

A greater electronegativity difference between bonded atoms generally increases bond polarity.

This can favour heterolytic cleavage when other conditions are suitable.

For example, the $\mathrm{C-Cl}$ bond is polarized as:

$$C\delta^{+} - Cl\delta^{-}$$

The electron pair is more likely to move towards chlorine than towards carbon during cleavage.

8.7.2 Polar Solvents

Polar solvents can stabilize ions through solvation.

Examples include:

- Water
- Methanol
- Ethanol
- Acetonitrile

Polar protic solvents can strongly solvate many ions, while polar aprotic solvents also influence ionic reactions through their dielectric properties and selective solvation.

8.7.3 Stability of the Resulting Ions

Heterolytic cleavage is favoured when the resulting ionic species are stabilized.

For example, carbocations may be stabilized by:

1. Resonance.

2. Hyperconjugation.

3. Electron-donating inductive effects.

4. Suitable neighbouring-group interactions.

Carbanions may be stabilized by:

1. Resonance.

2. Electron-withdrawing inductive effects.

3. Electron-withdrawing mesomeric effects.

4. Greater s-character of the orbital containing the lone pair.

8.7.4 Nature of the Leaving Group

A good leaving group can accept the bonding electron pair and depart relatively easily.

Common leaving groups include:

$$I^-, Br^-, Cl^-$$

Leaving-group ability depends on the solvent, substrate, and reaction mechanism.

8.8 Reactive Intermediates Formed During Heterolysis

Heterolytic cleavage can produce ionic intermediates, particularly carbocations and carbanions.

8.8.1 Carbocation

Definition: A carbocation is an organic ion in which a carbon atom carries a positive charge.

For example:

$$CH_3^+$$

Important Characteristics:

1. Carbocations are electron-deficient species.

2. Classical tricoordinate carbocations generally possess six valence electrons around the positively charged carbon.

3. The positively charged carbon is usually sp^2 hybridized.

4. It generally has a trigonal planar geometry.

5. It possesses a vacant p-orbital.

6. Carbocations generally behave as electrophiles.

$$CH_3^+$$

Carbocation geometry schematic

General Stability Order:

$$3° > 2° > 1° > CH_3^+$$

This order applies to simple alkyl carbocations.

8.8.2 Carbanion

Definition: A carbanion is an organic ion in which a carbon atom carries a negative charge and possesses a lone pair of electrons.

For example:

CH_3^-

Important Characteristics:

1. Carbanions possess a negatively charged carbon atom.
2. The carbon generally contains a lone pair of electrons.
3. Simple alkyl carbanions are often approximately trigonal pyramidal.
4. Many resonance-stabilized carbanions are planar or nearly planar.
5. Carbanions generally behave as nucleophiles and bases.
6. Their stability depends on resonance, inductive effects, hybridization, and solvation.

CH_3^-

Carbanion geometry schematic

General Stability Order for Simple Alkyl Carbanions:

$$CH_3^- > 1° > 2° > 3°$$

This trend reflects the destabilizing electron-donating effect of alkyl groups on simple negatively charged carbon centres.

However, resonance stabilization and other factors may change the order.

8.9 Heterolytic Reaction Mechanism (Ionic Mechanism)

A reaction mechanism involving the movement of electron pairs and the formation or participation of ionic species is commonly called an ionic mechanism.

Two important nucleophilic substitution mechanisms are:

- SN1 mechanism.
- SN2 mechanism.

8.9.1 SN1 Mechanism

SN1 stands for substitution nucleophilic unimolecular.

Example: Hydrolysis of tert-Butyl Chloride

$$(CH_3)_3CCl + H_2O \rightarrow (CH_3)_3COH + HCl$$

The mechanism proceeds through carbocation formation.

Stage 1: Ionization

$$(CH_3)_3CCl \rightarrow (CH_3)_3C^+ + Cl^-$$

This is the slow, rate-determining step.

Stage 2: Nucleophilic Attack

$$(CH_3)_3C^+ + H_2O \rightarrow (CH_3)_3COH_2^+$$

Stage 3: Deprotonation

$$(CH_3)_3COH_2^+ \rightarrow (CH_3)_3COH + H^+$$

The resulting proton is transferred to a suitable base in the reaction medium.

8.9.2 SN2 Mechanism

SN2 stands for substitution nucleophilic bimolecular.

Example: Reaction of Methyl Bromide with Hydroxide Ion

$$CH_3Br + OH^- \rightarrow CH_3OH + Br^-$$

$$\mathrm{OH^- + CH_3Br \rightarrow CH_3OH + Br^-}$$

SN2 substitution schematic

Explanation:

1. Hydroxide ion approaches the carbon atom from the side opposite the leaving group.
2. A new $\mathrm{C-O}$ bond begins to form.
3. Simultaneously, the $\mathrm{C-Br}$ bond begins to break.
4. Both bonding electrons of the $\mathrm{C-Br}$ bond move towards bromine.
5. Methanol and bromide ion are formed.

Important: SN2 is a concerted mechanism. A free carbocation is not formed.

8.10 Energy Required for Homolysis and Heterolysis

Bond cleavage requires energy, but the amount depends on the bond, the products formed, and the surrounding medium.

8.10.1 Energy of Homolytic Cleavage

The energy required to break a covalent bond homolytically in the gaseous state is related to its bond dissociation enthalpy.

For example:

$$Cl_2(g) \rightarrow 2Cl \bullet (g)$$

The bond dissociation enthalpy is approximately:

$$\Delta H \approx +243\ kJ\ mol^{-1}$$

8.10.2 Energy of Heterolytic Cleavage

Heterolytic cleavage produces ions.

For example:

$$A-B \rightarrow A^+ + B^-$$

In the gas phase, forming separated ions may require substantial energy because of charge separation and the energetic cost of ion formation.

However, in polar solvents, the resulting ions can be stabilized by solvation.

8.10.3 Is Heterolysis Always More Energy-Consuming Than Homolysis?

Heterolysis does not always require more energy than homolysis; the comparison depends on the bond and physical environment.

Explanation:

1. Gas-phase heterolysis of many ordinary covalent bonds requires considerably more energy than homolysis.

2. Heterolysis creates charged species whose energies depend strongly on their electronic structures.

3. Polar solvents can greatly stabilize the ionic products.

4. The nature of the leaving group and the stability of the ions also influence the energy requirement.

5. Therefore, the relative energy of homolysis and heterolysis depends on the chemical system and reaction conditions.

IIT-JEE Important Concept: Do not assume that heterolytic cleavage always requires more energy than homolytic cleavage. The comparison must specify the bond, products, and medium.

8.11 Comparison Between Homolytic and Heterolytic Bond Fission

Basis	Homolytic Fission	Heterolytic Fission
Nature of cleavage	Symmetrical electron division	Unsymmetrical electron division
Electron distribution	Each fragment receives one bonding electron	One fragment receives both bonding electrons
Species formed	Radicals	Ions
Charge on products	Often neutral radicals	Oppositely charged ions for a neutral bond
Favoured by	Suitable light, heat, radical initiators	Polar media, stable ions, good leaving groups
Electronegativity	Often small difference	Often significant bond polarization

Basis	Homolytic Fission	Heterolytic Fission
Arrow representation	single-barbed fish-hook arrows	full-headed curved arrow
Example	$Cl_2 \rightarrow 2Cl\bullet$	$A{-}B \rightarrow A^+ + B^-$
Typical reaction pathway	Free-radical mechanism	Ionic mechanism

8.12 Important IIT-JEE Examination Points

1. Bond fission is the breaking of a covalent bond.
2. Covalent bonds may undergo homolytic or heterolytic cleavage.
3. Homolysis produces fragments containing unpaired electrons.
4. Heterolysis involves transfer of both bonding electrons to one fragment.
5. Homolytic cleavage is represented by single-barbed fish-hook arrows.
6. Heterolytic cleavage is represented by full-headed curved arrows.
7. Heat, light, and radical initiators may promote homolytic cleavage.
8. Polar solvents often favour reactions involving ionic intermediates.
9. Free radicals are commonly formed through homolysis.
10. Carbocations and carbanions are important ionic intermediates.
11. Carbocations are generally electron deficient and behave as electrophiles.
12. Carbanions generally behave as nucleophiles and bases.
13. Simple alkyl carbocations usually follow the stability order 3° > 2° > 1° > CH_3^+.
14. Simple alkyl radicals generally follow the stability order 3° > 2° > 1° > $CH_3\bullet$.
15. Simple alkyl carbanions often show the opposite substitution trend to carbocations.
16. SN1 reactions involve carbocation formation in the rate-determining ionization step.
17. SN2 reactions occur through a concerted mechanism without formation of a free carbocation.
18. Free-radical chain reactions involve initiation, propagation, and termination stages.
19. The energy required for bond cleavage depends on the bond, resulting species, and reaction medium.
20. Heterolysis does not always require more energy than homolysis under every set of conditions.

9 Carbenes

9.1 Definition and electron count

A carbene is a neutral species containing a divalent carbon with two nonbonding electrons. Methylene, $:CH_2$, is the parent example; dichlorocarbene, $:CCl_2$, is another important example. In the simplest Lewis description, two σ bonds supply four electrons around the carbon and the two nonbonding electrons give a six-electron valence shell.

The carbene centre is therefore different from a classical carbocation, which has three bonds and no nonbonding pair, and from a carbanion, which usually has three bonds and a lone pair. The two nonbonding electrons of a carbene can occupy different spin states. Most elementary examples are short-lived intermediates, although specially stabilised carbenes can be isolated.

9.2 Singlet and triplet carbenes

A singlet carbene has total electron spin S = 0. In the common closed-shell description, its two nonbonding electrons are paired in one orbital and another orbital is vacant. It is diamagnetic. A bent, approximately sp^2 description is useful for many singlet carbenes: two orbitals make σ bonds, another contains a lone pair, and the p orbital is empty.

A triplet carbene has total spin S = 1, with two unpaired electrons in different orbitals, and is paramagnetic. It is also generally bent; do not describe every triplet carbene as obligatorily linear or assign a universal hybridisation and angle. Orbital energies, substituents and geometry determine the singlet–triplet separation.

Isolated methylene has a triplet ground state. Dichlorocarbene has a singlet ground state; chlorine lone-pair donation helps stabilise the vacant orbital arrangement of the singlet. The state produced in a reaction also depends on the precursor and reaction conditions. “All carbenes have a triplet ground state” is false.

9.3 Generation from diazo compounds

Loss of nitrogen from a diazo compound under appropriate photochemical or thermal conditions can generate a carbene. A simple schematic equation is $CH_2N_2 \rightarrow :CH_2 + N_2$, with light or heat written over the reaction arrow. The spin state and subsequent chemistry depend on the conditions and substituents.

A metal-catalysed diazo reaction may proceed through a metal carbenoid rather than a freely diffusing carbene. Distinguish the intermediate supported by the mechanism from a convenient formal label. These transformations describe reaction principles; diazo compounds are hazardous laboratory reagents and are not classroom handling exercises.

9.4 Dichlorocarbene from chloroform

A strong base removes the proton from chloroform to give trichloromethanide, which can eliminate chloride from the same carbon to generate dichlorocarbene. This is α elimination because proton removal and leaving-group loss involve the same carbon, unlike ordinary β elimination to form an alkene.

The electron and atom accounting is shown by $CHCl_3 + OH^- \rightleftharpoons CCl_3^- + H_2O$, followed by $CCl_3^- \rightarrow :CCl_2 + Cl^-$. The small equilibrium population of trichloromethanide can be consumed in the second step under suitable conditions. Overall: $CHCl_3 + OH^- \rightarrow :CCl_2 + Cl^- + H_2O$. Dichlorocarbene may be trapped by an alkene.

9.5 Addition to alkenes

A carbene or suitable carbenoid can add a carbon across an alkene π bond to form a cyclopropane ring. With $:CCl_2$ the added ring carbon carries both chlorine atoms, giving a gem-dichlorocyclopropane. Ethene plus $:CH_2$ formally gives cyclopropane; ethene plus $:CCl_2$ gives 1,1-dichlorocyclopropane.

Concerted singlet-carbene addition retains the relative arrangement of substituents on the alkene. Thus cis-2-butene gives a cis arrangement of its methyl groups in the cyclopropane product, whereas trans-2-butene gives a trans arrangement. “Stereospecific” here refers to this relationship; it does not imply that an achiral reaction must form only one enantiomer.

Triplet-carbene addition can be stepwise through a diradical. Bond rotation before ring closure can reduce stereospecificity. Product stereochemistry is mechanistic evidence, but avoid treating it as an unconditional rule independent of the lifetime and environment of the intermediate.

9.6 Insertion and rearrangement

Carbenes may insert into suitable $C-H$ or heteroatom–H bonds. A formal illustration is $:CH_2 + R-H \rightarrow R-CH_3$, where a new carbon–R bond and carbon–H bond are formed at the carbene carbon. Actual selectivity depends on the carbene, bond type and reaction conditions.

Some substituted carbenes undergo rapid 1,2 migration to form alkenes. In the Wolff rearrangement, loss of nitrogen from an α-diazoketone leads to a ketene through migration; the detailed pathway can depend on the substrate and conditions. Do not assume that every rearrangement requires a long-lived free carbene.

9.7 Carbenoids and the Simmons Smith reaction

A carbenoid is a metal-associated reagent or intermediate with carbene-like reactivity. It need not release a free carbene. In the Simmons–Smith reaction, CH_2I_2 with an appropriate zinc reagent forms an organozinc carbenoid, commonly represented as ICH_2ZnI, that transfers CH_2 to an alkene to give a cyclopropane.

This cyclopropanation usually retains the alkene substituent relationship. It is therefore useful to distinguish “cyclopropanation by transfer of a carbene unit” from a mechanism that specifically involves free $:CH_2$.

9.8 Comparison with other intermediates

Classical carbocation: commonly three σ bonds, six electrons around carbon, positive charge, and a vacant orbital. Simple carbanion: commonly three σ bonds, eight electrons around carbon, negative charge, and a lone pair. Simple carbon radical: commonly three σ bonds, seven electrons around carbon, and one unpaired electron. Carbene: two σ bonds, six electrons around carbon, no overall charge, and two nonbonding electrons.

These electron counts refer to the simple local Lewis models. Resonance, metal coordination and unusual bonding can require a more detailed description. A carbene is not a carbon radical merely because its triplet state has unpaired electrons.

9.9 Worked checks

Question: What intermediate forms when CCl_3^- loses Cl^-? Answer: Neutral $:CCl_2$. Charge is conserved because −1 = 0 + (−1).

Question: What ring forms when an alkene traps $:CCl_2$? Answer: A cyclopropane bearing two chlorines on the newly added carbon.

Question: Does observing a cyclopropane prove that free carbene was present? Answer: No. A metal carbenoid can transfer the same carbon unit.

Question: Which statement is reliable: all singlet carbenes are more stable, or stability depends on substitution? Answer: The latter. Triplet methylene and singlet dichlorocarbene illustrate why a universal order is wrong.

10 Nitrenes

10.1 Introduction

Nitrenes are highly reactive, short-lived nitrogen-containing reaction intermediates. They are considered the nitrogen analogues of carbenes.

Nitrenes are nitrogen-centred reactive intermediates whose structure and reactivity depend strongly on their electronic state.

10.1.1 Definition

A nitrene RN is a neutral monovalent nitrogen species with six electrons around nitrogen in the simple Lewis model: two in the R – N bond and four nonbonding electrons. In the elementary spin-state description, one nonbonding pair remains paired while the other two electrons distinguish the singlet and triplet arrangements.

Its general representation is:

$$R - N:$$

Here, R may represent an alkyl, aryl, or another suitable substituent.

Important Points:

- Nitrenes are the nitrogen analogues of carbenes.
- The nitrogen atom is bonded to only one substituent.
- Nitrenes are neutral and electron-deficient reactive intermediates.
- They possess four nonbonding electrons in total: one ordinary lone pair and two additional nonbonding electrons.
- The two additional electrons may be paired or unpaired.
- Nitrenes can exist in singlet and triplet electronic states.
- They are generally highly reactive and are often generated in situ, meaning within the reaction mixture.

10.2 Electronic Structure of Nitrenes

Nitrogen has the electronic configuration:

$$1s^2\ 2s^2\ 2p^3$$

It possesses five valence electrons.

In a nitrene, nitrogen forms one covalent bond with a substituent R.

The nitrogen atom therefore has:

- One σ-bond with R.
- One ordinary lone pair of electrons.
- Two additional nonbonding electrons.

10.2.1 Electron Arrangement

A simplified representation is:

$$R-N\colon\ (one\ ordinary\ lone\ pair\ +\ two\ additional\ nonbonding\ electrons)$$

Conceptual electron-dot representation of a nitrene. The two additional electrons are shown separately to emphasize their possible spin arrangements.

Explanation:

- The nitrogen atom forms a σ-bond with the substituent.
- One pair of electrons remains as an ordinary lone pair.
- Two additional electrons occupy nonbonding orbitals.
- Their arrangement determines whether the nitrene is singlet or triplet.

10.3 Types of Nitrenes

Nitrenes are classified into two electronic states:

- Singlet nitrene
- Triplet nitrene

The distinction depends on the spin arrangement of the two additional nonbonding electrons.

10.3.1 Singlet Nitrene

Definition

A singlet nitrene is a nitrene in which the two additional nonbonding electrons have opposite spins and are paired.

The total electronic spin is:

$$S\ =\ 0$$

The spin multiplicity is:

$$2S\ +\ 1\ =\ 1$$

Therefore, it is called a singlet nitrene.

Electronic Arrangement

$$Lone\ pair\colon\ \uparrow\downarrow \quad Nonbonding\colon\ \uparrow\downarrow \quad Vacant\ orbital\colon [\ ]$$

Simplified orbital occupation for a singlet nitrene

Important Characteristics:

- The two additional nonbonding electrons are paired.
- The total spin quantum number is zero.
- Singlet nitrenes are diamagnetic in the idealized closed-shell description.
- An empty orbital may be available for accepting electron density.
- Singlet nitrenes can undergo concerted addition and insertion reactions.

IIT-JEE Important Concept: Singlet nitrenes can exhibit electrophilic behaviour because of their electron-deficient nitrogen centre.

10.3.2 Triplet Nitrene

Definition

A triplet nitrene is a nitrene in which the two additional nonbonding electrons occupy different orbitals with parallel spins.

The total electronic spin is:

$$S = 1$$

The spin multiplicity is:

$$2S + 1 = 3$$

Therefore, it is called a triplet nitrene.

Electronic Arrangement

$$Lone\ pair: \uparrow\downarrow \quad Orbital\ 1: \uparrow \quad Orbital\ 2: \uparrow$$

Simplified orbital occupation for a triplet nitrene

Important Characteristics:

- The two additional nonbonding electrons are unpaired.
- They occupy different orbitals.
- Their spins are parallel, consistent with Hund's rule when the relevant orbitals are degenerate or nearly degenerate.
- The total spin quantum number is one.
- Triplet nitrenes are paramagnetic.
- Their reactions may involve stepwise radical-like pathways.

10.3.3 Ground-State Stability

Triplet NH is the ground state of the parent nitrene. Substitution changes the singlet–triplet energy separation.

For simple parent nitrenes such as NH, the triplet state is the ground state.

However, this is not a universal rule for every substituted nitrene. Substituents and electronic delocalization can change the relative energies of singlet and triplet states.

$$Triplet\ state\ is\ the\ ground\ state\ of\ parent\ NH$$

10.4 Comparison Between Singlet and Triplet Nitrenes

Characteristic	Singlet Nitrene	Triplet Nitrene
Additional nonbonding electrons	Paired	Unpaired
Electron spins	Opposite	Parallel

Characteristic	Singlet Nitrene	Triplet Nitrene
Total spin S	0	1
Spin multiplicity	1	3
Magnetic nature	Diamagnetic	Paramagnetic
Orbital occupation	Two electrons paired in one orbital in the simple model	Two electrons occupy separate orbitals
Typical reaction behaviour	Often concerted	Often stepwise, radical-like
Ground-state preference	Depends on substitution	Ground state for parent NH

10.5 Hybridization and Structure of Nitrenes

The structure of a nitrene depends on its electronic state and substituents.

Simple orbital pictures can aid electron counting, but exact geometry and orbital energies depend on the specific nitrene.

Explanation:

- Nitrogen forms one σ-bond with R.
- One lone pair remains paired.
- The two additional electrons may occupy different nonbonding orbitals.
- In the simple sp-hybridized model, the two remaining p-orbitals can accommodate one electron each.
- These electrons have parallel spins in the triplet state.

Important: Hybridization is a simplified bonding model. The actual orbital character and molecular geometry depend on the particular nitrene and should not be assumed identical for all nitrenes.

10.6 Generation of Nitrenes and Related Rearrangements

Azide decomposition is a standard route to nitrenes. The Hofmann rearrangement is discussed separately here as a related nitrogen-centred rearrangement.

- Thermal or photochemical decomposition of suitable organic azides can generate nitrenes.
- The Hofmann bromamide reaction converts a primary amide into an amine with one fewer carbon atom.
- This rearrangement must not be assumed to require an isolable or freely diffusing nitrene.

10.6.1 Related Rearrangement The Hofmann Bromamide Reaction

Definition

The Hofmann bromamide reaction converts a primary amide into a primary amine containing one carbon atom fewer than the starting amide.

The reaction uses bromine and a strong base such as sodium hydroxide.

General Reaction

$$RCONH_2 + Br_2 + 4NaOH \rightarrow RNH_2 + Na_2CO_3 + 2NaBr + 2H_2O$$

For example, acetamide produces methylamine:

$$CH_3CONH_2 \rightarrow (Br_2/NaOH)\ CH_3NH_2$$

Explanation

- The primary amide reacts with bromine in a basic medium.
- An N-bromoamide intermediate is formed.
- Deprotonation produces a species capable of rearrangement.
- The organic group migrates from the carbonyl carbon to nitrogen.
- An isocyanate intermediate is formed.
- Hydrolysis and decarboxylation of the isocyanate produce the primary amine.

Reaction Sequence

$$Primary\ amide\ \ RCONH_2$$
$$\downarrow\ Br_2\ /\ OH^-$$
$$N-Bromoamide\ \ RCONHBr$$
$$\downarrow\ Deprotonation\ and\ rearrangement$$
$$Isocyanate\ \ RN=C=O$$
$$\downarrow\ Hydrolysis\ and\ decarboxylation$$
$$Primary\ amine\ \ RNH_2$$

Does the Hofmann Reaction Produce a Free Nitrene?

Traditional textbook explanations sometimes describe the rearrangement through a nitrene-like intermediate.

However, the commonly accepted Hofmann rearrangement mechanism involves migration to nitrogen with loss of bromide, without requiring a freely existing nitrene.

IIT-JEE Important Concept: The important intermediate in the conventional Hofmann bromamide mechanism is the isocyanate. A free nitrene should not be assumed to form as a necessary intermediate.

10.6.2 Decomposition of Organic Azides

Organic azides have the general formula:

RN_3

On heating or exposure to suitable light, some organic azides decompose with elimination of nitrogen gas.

General Reaction

$$RN_3 \rightarrow (h\nu \text{ or } \Delta)\ RN{:} + N_2$$

Here:

- RN_3 is an organic azide.
- RN: represents a nitrene.
- N_2 is nitrogen gas.

Explanation

- The organic azide absorbs thermal or photochemical energy.
- The azide group undergoes decomposition.
- Molecular nitrogen is eliminated.
- A nitrene or a closely related reactive species may be generated.
- The nitrene subsequently undergoes addition, insertion, or rearrangement reactions.

Important: Not every organic azide produces an isolable or freely reacting nitrene. Some azides undergo concerted rearrangements or other pathways.

10.6.3 Formation of Unsubstituted Nitrene

The unsubstituted nitrene is called imidogen, with the formula:

NH

It is a highly reactive species containing one $N-H$ bond.

Methods of Generation

The parent NH species can be generated in specialised gas-phase photochemical or discharge experiments with suitable nitrogen-containing precursors.

For conceptual understanding:

- Suitable nitrogen–hydrogen precursors may generate NH under photochemical conditions.
- Electrical discharge can produce reactive nitrogen-containing fragments in appropriate gaseous mixtures.
- The resulting NH species is generally short-lived.
- Its ground electronic state is triplet.

The product distribution and electronic state depend on the precursor and experimental conditions.

10.7 Chemical Reactions of Nitrenes

The following transformations illustrate typical modes of nitrene reactivity.

10.7.1 Addition to Alkenes

Nitrenes can add to carbon–carbon double bonds to form three-membered nitrogen-containing rings called aziridines.

A simplified reaction is:

$$R - N\colon + \ CH_2 = CH_2 \ \rightarrow \ N - substituted\ aziridine$$

Simplified structure of an N-substituted aziridine: three-membered N–CH_2–CH_2 ring with R attached to nitrogen.

Explanation:

- The nitrene interacts with the π-bond of an alkene.
- Two new $\mathrm{C} - \mathrm{N}$ bonds form.
- A three-membered ring is produced.
- The product is called an aziridine.

Singlet nitrenes may undergo concerted addition, whereas triplet nitrenes may react through stepwise pathways.

10.7.2 Insertion into $\mathbf{C} - \mathbf{H}$ Bonds

Nitrenes can insert into suitable carbon–hydrogen bonds.

A simplified representation is:

$$RN\colon + \ R'H \ \rightarrow \ RNH - R'$$

Explanation:

- The nitrene approaches a $\mathrm{C} - \mathrm{H}$ bond.
- A new $\mathrm{C} - \mathrm{N}$ bond is formed.
- A new $\mathrm{N} - \mathrm{H}$ bond is also formed.
- The overall process introduces nitrogen into the organic molecule.

10.7.3 Rearrangement Reactions

Nitrenes or nitrene-like species may undergo rearrangement through migration of an adjacent group.

Such reactions are important in understanding nitrogen insertion and skeletal rearrangement chemistry.

However, the exact mechanism depends on the substrate, and a free nitrene is not required in every reaction traditionally described as involving nitrene-like rearrangement.

10.8 Comparison Between Carbenes and Nitrenes

Characteristic	Carbene	Nitrene
Central atom	Carbon	Nitrogen

Characteristic	Carbene	Nitrene
General representation	R_2C:	RN:
Covalent bonds to substituents	Two	One
Nature	Neutral reactive intermediate	Neutral reactive intermediate
Electronic states	Singlet and triplet	Singlet and triplet
Magnetic nature	Singlet diamagnetic, triplet paramagnetic	Singlet diamagnetic, triplet paramagnetic
Common generation method	Decomposition of diazo compounds or other suitable precursors	Decomposition of organic azides
Addition to alkene	May form cyclopropanes	May form aziridines
Insertion reactions	Possible	Possible

10.9 Important IIT-JEE Examination Points

- Nitrenes are the nitrogen analogues of carbenes.
- The general formula of a nitrene is RN:.
- Nitrenes are neutral, monovalent nitrogen-containing reactive intermediates.
- Nitrenes possess one ordinary lone pair and two additional nonbonding electrons.
- The two additional electrons may be paired or unpaired.
- Singlet nitrenes have paired additional nonbonding electrons.
- Triplet nitrenes have two unpaired electrons with parallel spins.
- Singlet nitrenes are diamagnetic in the simple closed-shell model.
- Triplet nitrenes are paramagnetic.
- The spin multiplicity of a singlet nitrene is one.
- The spin multiplicity of a triplet nitrene is three.
- Parent NH has a triplet ground state.
- The relative stability of singlet and triplet nitrenes depends on substituents.
- Nitrenes can be generated by thermal or photochemical decomposition of suitable organic azides.
- The decomposition of organic azides may involve elimination of molecular nitrogen.
- Nitrenes can undergo addition to alkenes to form aziridines.
- Nitrenes may undergo insertion into suitable $C-H$ bonds.

- The Hofmann bromamide reaction converts a primary amide into a primary amine with one carbon atom fewer.
- The conventional Hofmann rearrangement proceeds through an isocyanate intermediate.
- A freely existing nitrene is not required in the accepted Hofmann rearrangement mechanism.
- The unsubstituted nitrene is called imidogen, NH.
- Nitrenes are often generated in situ for nitrogen transfer reactions.

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11 Electrophiles and nucleophiles

11.1 Introduction

During an organic chemical reaction, existing bonds may break and new bonds may form. These changes are often initiated by chemical species known as attacking reagents.

An attacking reagent interacts with an electron-rich or electron-deficient centre of a substrate molecule and brings about a chemical reaction.

11.1.1 Definition

An attacking reagent is a chemical species that attacks a suitable reactive centre in a substrate and participates in bond formation or bond breaking.

The attacking reagent may be positively charged, negatively charged, or electrically neutral.

11.1.2 Classification of Attacking Reagents

Most attacking reagents are classified into two main categories:

- Electrophiles (Electrophilic Reagents) — electron-pair acceptors.
- Nucleophiles (Nucleophilic Reagents) — electron-pair donors.

$$Attacking\ Reagents \rightarrow Electrophiles\ (electron-pair\ acceptors) + Nucleophiles\ (electron-pair\ donors)$$

Important Points

- Electrophiles attack electron-rich centres.
- Nucleophiles attack electron-deficient centres.
- Electrophiles accept electron pairs and behave as Lewis acids.
- Nucleophiles donate electron pairs and behave as Lewis bases.
- The reaction between an electrophile and a nucleophile often results in the formation of a new covalent bond.

11.2 Electrophiles (Electrophilic Reagents)

11.2.1 Definition

An electrophile is an electron-deficient chemical species that accepts an electron pair from another species to form a covalent bond.

The word electrophile means electron-loving. Electrophiles are also called Lewis acids because they accept electron pairs.

General Representation

$$E^+ + :Nu^- \rightarrow E-Nu$$

Here:

- E^+ represents a positively charged electrophile.
- Nu^- represents a negatively charged nucleophile.
- The nucleophile donates an electron pair to the electrophile.

The equation is a general illustration; electrophiles and nucleophiles need not always carry charges.

11.2.2 Characteristics of Electrophiles

- Electrophiles are electron-pair acceptors.
- They are generally electron deficient.
- They may be positively charged or electrically neutral.
- They attack regions of high electron density.
- They may possess vacant orbitals or suitable electron-accepting orbitals.
- Electrophiles behave as Lewis acids.
- Their electrophilic character depends on electronic structure, substituents, solvent, and reaction conditions.

11.3 Classification of Electrophiles

Electrophiles are broadly classified into:

- Charged electrophiles
- Neutral electrophiles

11.3.1 Charged Electrophiles

Definition

Positively charged species capable of accepting an electron pair are called charged electrophiles. Many charged electrophiles contain an electron-deficient atom or a positively charged centre.

Examples

$$H^+, NO_2{}^+, R_3C^+, RCO^+$$

Electrophile	Name	Important Feature
H^+	Proton	Accepts an electron pair
$\mathbf{NO_2{}^+}$	Nitronium ion	Electrophile in aromatic nitration
$\mathbf{R_3C^+}$	Carbocation	Electron-deficient carbon with a vacant orbital

Electrophile	Name	Important Feature
$\mathbf{RCO^+}$	Acylium ion	Resonance-stabilized electrophile in acylation

Example: Protonation of Ammonia

Ammonia possesses a lone pair of electrons on nitrogen. The proton accepts this electron pair.

$$NH_3 + H^+ \rightarrow NH_4^+$$

Explanation:

- Ammonia acts as an electron-pair donor.
- The proton acts as an electron-pair acceptor.
- Nitrogen donates its lone pair towards the proton.
- A new $\mathrm{N-H}$ bond is formed.
- Ammonium ion is produced.

Important Clarification

Metal cations can act as Lewis acids, but their coordination chemistry should be distinguished from the usual organic electrophilic-substitution examples.

For IIT-JEE, the more reliable criterion is electron-pair accepting ability. Not every cation behaves as an electrophile in the same way, and metal cations can also act as Lewis acids by accepting electron density from ligands.

11.3.2 Neutral Electrophiles

Definition

Electrically neutral species capable of accepting an electron pair are called neutral electrophiles. They include electron-deficient molecules and molecules with polarised multiple bonds.

(i) Neutral Compounds Having an Incomplete Octet

Some neutral compounds possess a central atom with fewer than eight electrons. Such compounds can accept an electron pair from a suitable donor.

Examples:

$$BF_3, BCl_3, AlCl_3$$

Example: Boron Trifluoride

In $\mathrm{BF_3}$, boron forms three covalent bonds and has only six electrons around it. It possesses an empty orbital capable of accepting an electron pair.

$$BF_3 + NH_3 \rightarrow F_3B \leftarrow NH_3$$

Explanation:

- Boron trifluoride is electron deficient.
- The boron atom possesses an available acceptor orbital.
- Ammonia contains a lone pair on nitrogen.
- Nitrogen donates its electron pair to boron.
- A Lewis acid–base adduct is formed.

Therefore, BF_3 acts as an electrophile and NH_3 acts as a nucleophile.

(ii) Neutral Compounds Capable of Accepting Electron Density Through Available Orbitals

Some neutral compounds can accept electron density even though the central atom may have a complete or expanded valence shell in a conventional Lewis structure.

Examples:

$$SiCl_4, PCl_5$$

Electrophilic behaviour is explained by suitable accepting orbitals and bond polarisation; it should not be attributed automatically to vacant d orbitals.

Important clarification: Modern bonding descriptions do not generally require significant vacant d-orbital participation to explain their Lewis acidity. Their electron-accepting behaviour is better understood using suitable molecular orbitals and bond polarization.

(iii) Neutral Molecules Containing Strongly Polarized Bonds

Some neutral compounds act as electrophiles because a particular atom carries a partial positive charge.

$$C\delta^+ = O\delta^-$$

The carbonyl carbon is electron deficient. Therefore, aldehydes and ketones can undergo nucleophilic attack at the carbonyl carbon.

Other examples: Certain alkyl halides, acid derivatives, and suitably activated multiple-bond compounds also contain electrophilic centres.

Important Concept

A neutral molecule does not need to possess an incomplete octet to behave as an electrophile. A sufficiently electron-deficient reaction centre may accept an electron pair.

11.4 Nucleophiles (Nucleophilic Reagents)

11.4.1 Definition

A nucleophile is an electron-rich chemical species that donates an electron pair to an electrophile to form a covalent bond.

The word nucleophile means nucleus-loving. Nucleophiles are also called Lewis bases because they donate electron pairs.

General Representation

$$Nu^- + E^+ \rightarrow Nu - E$$

11.4.2 Characteristics of Nucleophiles

- Nucleophiles are electron-pair donors.
- They generally possess lone pairs, negative charges, or π-electrons.
- They attack electron-deficient centres.
- They may be negatively charged or electrically neutral.
- Nucleophiles behave as Lewis bases.
- Their nucleophilic strength depends on electronic structure, solvent, steric effects, and the electrophile involved.
- Many nucleophiles are also Brønsted bases, although nucleophilicity and basicity are not identical properties.

11.5 Classification of Nucleophiles

Nucleophiles are classified into three important categories:

- Charged nucleophiles
- Neutral nucleophiles
- Ambident nucleophiles

11.5.1 Charged Nucleophiles

Definition

Negatively charged species capable of donating an electron pair are called charged nucleophiles. They generally possess high electron density and can form covalent bonds with electrophilic centres.

Important Examples

Nucleophile	Name	Attacking Atom
OH^-	Hydroxide ion	Oxygen
RO^-	Alkoxide ion	Oxygen
CN^-	Cyanide ion	Carbon or nitrogen
NH_2^-	Amide ion	Nitrogen
Cl^-	Chloride ion	Chlorine
Br^-	Bromide ion	Bromine
I^-	Iodide ion	Iodine

Nucleophile	Name	Attacking Atom
$\mathbf{RS^-}$	Thiolate ion	Sulfur

Example: Reaction of Methyl Bromide with Hydroxide Ion

$$CH_3Br + OH^- \rightarrow CH_3OH + Br^-$$

Explanation:

- The $\mathrm{C-Br}$ bond is polarized because bromine is more electronegative than carbon.
- Carbon acquires a partial positive charge.
- Hydroxide ion acts as a nucleophile.
- It attacks the carbon atom and forms a new $\mathrm{C-O}$ bond.
- Simultaneously, the $\mathrm{C-Br}$ bond breaks and bromide ion leaves.
- Methanol is formed.

This reaction follows an SN2 mechanism.

11.5.2 Neutral Nucleophiles

Definition

Neutral molecules with an available lone pair or an electron-rich pi system can behave as nucleophiles.

(i) Neutral Molecules Containing Lone Pairs

Neutral compounds containing atoms such as nitrogen, oxygen, sulfur, or phosphorus may donate lone pairs to electrophiles.

Examples:

Type	Examples	Electron-Donating Atom
Nitrogen nucleophiles	$\mathrm{NH_3}$, $\mathrm{RNH_2}$, $\mathrm{R_2NH}$	Nitrogen
Oxygen nucleophiles	$\mathrm{H_2O}$, ROH, $\mathrm{R_2O}$	Oxygen
Sulfur nucleophiles	$\mathrm{H_2S}$, RSH, $\mathrm{R_2S}$	Sulfur
Phosphorus nucleophiles	$\mathrm{PR_3}$	Phosphorus

Example: Ammonia as a Nucleophile

Consider the reaction of ammonia with methyl bromide.

$$CH_3Br + NH_3 \rightarrow CH_3NH_3^+ + Br^-$$

Subsequent deprotonation produces methylamine:

$$CH_3NH_3^+ \rightarrow CH_3NH_2 + H^+$$

The proton is accepted by a base present in the reaction mixture.

Explanation:

- Ammonia contains a lone pair of electrons on nitrogen.
- The nitrogen atom attacks the electrophilic carbon.
- A new $C-N$ bond forms.
- Bromide ion leaves.
- Deprotonation gives methylamine.

(ii) Neutral Organic Compounds Containing π-Bonds

Organic compounds containing carbon–carbon multiple bonds can act as nucleophiles because their π-electrons are available for interaction with electrophiles.

Important Examples:

- Alkenes
- Alkynes
- Benzene and other aromatic compounds

Example: Electrophilic Addition to Ethene

$$CH_2 = CH_2 + H^+ \rightarrow CH_3CH_2^+$$

Explanation:

- Ethene contains a π-bond.
- The π-electrons act as an electron source.
- The proton acts as an electrophile.
- A new $C-H$ bond forms.
- An ethyl carbocation is produced.

The carbocation may subsequently react with a nucleophile.

Example: Electrophilic Substitution in Benzene

Benzene contains a delocalized π-electron system. During nitration, the aromatic π-system attacks the nitronium ion.

$$C_6H_6 + NO_2^+ \rightarrow arenium\ ion \rightarrow C_6H_5NO_2 + H^+$$

The final product is nitrobenzene:

$$C_6H_5NO_2$$

Important Concept: Benzene can act as a nucleophile towards sufficiently strong electrophiles, although its aromatic stabilization makes it less reactive towards ordinary electrophilic addition than typical alkenes.

11.6 Ambident Nucleophiles

11.6.1 Definition

An ambident nucleophile is a nucleophilic species that can attack an electrophile through two different atoms or reactive centres.

The two sites may have different electronic environments and may lead to different products.

Important Points

- Ambident nucleophiles possess two potential nucleophilic centres.
- They can form bonds through either attacking site.
- The products formed depend on the electrophile, solvent, reaction conditions, and nucleophile structure.
- Ambident nucleophiles are important in understanding regioselectivity and product formation.

11.6.2 Examples of Ambident Nucleophiles

Ambident Nucleophile	Possible Attacking Sites
$\mathbf{CN^-}$	Carbon and nitrogen
$\mathbf{NO_2^-}$	Nitrogen and oxygen
Enolate ion	Carbon and oxygen
$\mathbf{SCN^-}$	Sulfur and nitrogen

11.6.3 Example: Cyanide Ion

The cyanide ion can attack through carbon or nitrogen. Its two possible modes of attachment can lead to different organic products.

For an alkyl halide $\mathrm{R-X}$:

Attack through carbon:

$$R - X + CN^- \rightarrow R - CN + X^-$$

The product is an alkyl cyanide or nitrile.

Attack through nitrogen:

$$R - X + CN^- \rightarrow R - NC + X^-$$

The product is an alkyl isocyanide.

11.6.4 Example: Nitrite Ion

Nitrite ion can attack through nitrogen or oxygen.

Attack through nitrogen:

$$R - X + NO_2^- \rightarrow R - NO_2 + X^-$$

The product is a nitroalkane.

Attack through oxygen:

$$R - X + NO_2^- \rightarrow R - O - N = O + X^-$$

The product is an alkyl nitrite.

IIT-JEE Important Concept: Ambident nucleophiles can produce different constitutional isomers depending on the atom through which they attack.

11.7 Organometallic Compounds as Nucleophiles

Many organometallic reagents contain a polarised carbon–metal bond and behave as carbon nucleophiles.

$$\mathrm{R - MgX}$$

where R–MgX represents a Grignard reagent.

11.7.1 Nature of the Carbon–Magnesium Bond

Magnesium is more electropositive than carbon. Therefore, the carbon–magnesium bond is polarized as:

$$R\delta^- - MgX\delta^+$$

The carbon atom has carbanion-like character and can attack electrophilic centres.

11.7.2 Example: Addition of a Grignard Reagent to Formaldehyde

$$HCHO + RMgX \rightarrow RCH_2O^-MgX^+$$

On hydrolysis:

$\mathrm{RCH_2O^- + H_3O^+ \rightarrow RCH_2OH + H_2O}$

This balanced net ionic equation shows protonation of the alkoxide; magnesium-containing counterions are omitted.

Explanation:

- The carbon atom bonded to magnesium behaves as a nucleophilic centre.
- It attacks the electrophilic carbonyl carbon.
- The $\mathrm{C = O}$ π-electrons move towards oxygen.
- A magnesium alkoxide intermediate forms.
- Acidic work-up produces an alcohol.

Important: Not all organometallic compounds behave identically. Their nucleophilicity depends on the metal, carbon–metal bond, ligands, and reaction conditions.

11.8 Organic Compounds Acting as Both Electrophiles and Nucleophiles

Some molecules contain both electron-deficient and electron-rich centres. Therefore, different atoms within the same molecule can participate in electrophilic and nucleophilic interactions.

11.8.1 Example: Carbonyl Compounds

Consider an aldehyde or ketone:

$$R_2C = O$$

The $C = O$ bond is polarized:

$$C\delta^+ = O\delta^-$$

Electrophilic centre:

The carbonyl carbon carries a partial positive charge and can accept an electron pair from a nucleophile.

Nucleophilic centre:

The oxygen atom contains lone pairs and can donate an electron pair to a suitable electrophile, such as a proton.

Example: Protonation of a Carbonyl Compound

$$R_2C = O + H^+ \rightarrow R_2C = OH^+$$

Here, oxygen acts as an electron-pair donor.

Example: Nucleophilic Addition to a Carbonyl Compound

$$R_2C = O + CN^- \rightarrow R_2C(CN)O^-$$

Here, carbonyl carbon acts as the electrophilic centre.

Important Points

- A molecule may contain more than one reactive centre.
- The carbonyl carbon is generally electrophilic.
- Carbonyl oxygen can act as a nucleophilic or basic centre.
- The type of reaction depends on the attacking reagent.
- Similar considerations may apply to other polar multiple-bond systems containing nitrogen or sulfur.

11.9 Relationship Between Electrophiles, Nucleophiles, Lewis Acids and Lewis Bases

11.9.1 Electrophiles Are Lewis Acids

A Lewis acid is a species that accepts an electron pair. Therefore, electrophiles are Lewis acids in the context of bond formation.

Examples:

$$BF_3, AlCl_3, H^+, NO_2{}^+$$

11.9.2 Nucleophiles Are Lewis Bases

A Lewis base is a species that donates an electron pair. Therefore, nucleophiles are Lewis bases.

Examples:

$$OH^-, NH_3, H_2O, CN^-$$

11.9.3 General Lewis Acid–Base Reaction

$$Lewis\ Acid\ +\ Lewis\ Base\ \rightarrow\ Adduct$$

For example:

$$BF_3\ +\ NH_3\ \rightarrow\ F_3B \leftarrow NH_3$$

11.9.4 Relationship Between Acid Strength and Conjugate Base Strength

For a Brønsted acid–base conjugate pair:

$$HA\ \rightleftharpoons\ H^+\ +\ A^-$$

A stronger acid generally has a weaker conjugate base.

Important: A weaker conjugate base is not necessarily a weaker nucleophile under all conditions.

Nucleophilicity is a kinetic property, whereas basicity is a thermodynamic property. Solvent effects and polarizability can produce different orders.

11.10 Nucleophilicity and Basicity

Although nucleophiles and bases both donate electron pairs, their roles are different.

11.10.1 Basicity

Basicity describes the tendency of a species to accept a proton.

$$OH^-\ +\ H^+\ \rightarrow\ H_2O$$

11.10.2 Nucleophilicity

Nucleophilicity describes the tendency of a species to attack an electrophilic centre and form a bond.

$$OH^-\ +\ CH_3Br\ \rightarrow\ CH_3OH\ +\ Br^-$$

11.10.3 Difference Between Basicity and Nucleophilicity

Basis	Basicity	Nucleophilicity
Nature	Thermodynamic tendency	Kinetic reactivity
Attacking centre	Proton	Electrophilic atom
Measured through	Equilibrium constants, pKa	Reaction rates
Strongly affected by	Stability of conjugate acid/base	Solvent, steric effects, polarizability

Basis	Basicity	Nucleophilicity
Example	OH^- accepting H^+	OH^- attacking CH_3Br

11.11 Factors Affecting Nucleophilicity

11.11.1 Negative Charge

For closely related species, a negatively charged nucleophile is often more reactive than its neutral conjugate acid.

$$OH^- > H_2O$$

in many comparable nucleophilic reactions.

11.11.2 Electronegativity

Across a period, nucleophilicity often decreases as electronegativity increases when comparing similarly structured nucleophiles under comparable conditions. However, solvent and other effects must also be considered.

11.11.3 Solvent Effect

In polar protic solvents, smaller anions may be strongly solvated through hydrogen bonding.

For halide ions, a commonly observed nucleophilicity order in protic solvents is:

$$I^- > Br^- > Cl^- > F^-$$

In suitable polar aprotic solvents, the order may be reversed:

$$F^- > Cl^- > Br^- > I^-$$

These are common trends rather than universal rankings for every reaction system.

11.11.4 Steric Hindrance

Bulky nucleophiles may have difficulty approaching a crowded electrophilic carbon. For example, tert-butoxide is a strong base but is relatively hindered as an SN2 nucleophile.

11.11.5 Polarizability

Highly polarizable nucleophiles can interact effectively with suitable electrophilic centres. This is one reason larger, softer nucleophiles may be particularly reactive in certain substitution reactions.

11.12 Comparison Between Electrophiles and Nucleophiles

Characteristic	Electrophile	Nucleophile
Meaning	Electron-loving	Nucleus-loving

Characteristic	Electrophile	Nucleophile
Electron-pair role	Acceptor	Donor
Electronic nature	Electron deficient	Electron rich
Lewis classification	Lewis acid	Lewis base
Common charge	Positive or neutral	Negative or neutral
Attacking region	Electron-rich centre	Electron-deficient centre
Examples	H^+, $NO_2{}^+$, BF_3, carbocations	OH^-, CN^-, NH_3, H_2O
Important reaction	Electrophilic addition or substitution	Nucleophilic addition or substitution

11.13 Important IIT-JEE Examination Points

- Attacking reagents are broadly classified into electrophiles and nucleophiles.
- Electrophiles are electron-pair acceptors.
- Nucleophiles are electron-pair donors.
- Electrophiles behave as Lewis acids.
- Nucleophiles behave as Lewis bases.
- Electrophiles may be positively charged or electrically neutral.
- Nucleophiles may be negatively charged or electrically neutral.
- Carbocations are important examples of charged electrophiles.
- The nitronium ion, $NO_2{}^+$, acts as an electrophile in aromatic nitration.
- BF_3 and $AlCl_3$ are important Lewis acids.
- Neutral molecules such as ammonia, water, alcohols, amines, and phosphines may act as nucleophiles.
- Alkenes, alkynes, and aromatic compounds can behave as nucleophiles through their π-electrons.
- Hydroxide, alkoxide, cyanide, and halide ions are common negatively charged nucleophiles.
- Ambident nucleophiles possess two possible attacking centres.
- Cyanide, nitrite, thiocyanate, and enolate ions are important examples of ambident nucleophiles.
- Cyanide ion may form nitriles or isocyanides depending on its mode of attack and reaction conditions.
- Nitrite ion may form nitroalkanes or alkyl nitrites.

- Organometallic compounds such as Grignard reagents can behave as carbon nucleophiles.
- Carbonyl compounds contain an electrophilic carbon atom and an electron-rich oxygen atom.
- A molecule may contain both electrophilic and nucleophilic centres.
- Basicity is a thermodynamic property, whereas nucleophilicity is a kinetic property.
- A strong base is not necessarily a strong nucleophile in every reaction.
- Solvent polarity, hydrogen bonding, steric hindrance, and polarizability influence nucleophilicity.
- In polar protic solvents, the common halide nucleophilicity order is $I^- > Br^- > Cl^- > F^-$.
- In suitable polar aprotic solvents, the halide nucleophilicity order may be reversed.
- Electron-pair movement in organic mechanisms is represented by curved arrows beginning at an electron pair or bond.
- During a typical electrophile–nucleophile reaction, the nucleophile supplies the electron pair used to form the new bond.

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12 Integrated GOC concepts

12.1 Rules for drawing resonance contributors

Resonance contributors have the same arrangement of nuclei and the same total charge and number of electrons. Only electron placement changes. Move π electrons or suitable lone pairs; do not move atoms or break the σ skeleton to turn one ordinary resonance contributor into another. Equivalent contributors carry equal weight. The actual species is one delocalised structure, not molecules rapidly changing into one another.

For second-period atoms, avoid drawing more than an octet. In common comparisons, contributors with complete octets, more bonding and less unnecessary charge separation are often important. Negative formal charge is usually better accommodated on a more electronegative atom, but this guideline must be considered alongside octet completion and aromaticity. A minor charge-separated contributor may still help explain reactivity.

For acetate, $CH_3 - C(=O) - O^- \leftrightarrow CH_3 - C(-O^-) = O$. The two oxygen atoms are equivalent in the isolated resonance hybrid; the negative charge is delocalised. For an amide, donation from nitrogen produces a contributor with N^+ and O^-. This explains restricted $C - N$ rotation and the reduced availability of the nitrogen lone pair.

12.2 Aromaticity as a source of stabilisation

For a simple monocyclic conjugated system, aromaticity requires a cyclic, approximately planar, continuously overlapping array of p orbitals containing (4n + 2) π electrons, with n a nonnegative integer. A planar, fully conjugated monocycle with 4n π electrons is antiaromatic. If the ring is not continuously conjugated or is sufficiently nonplanar, it is nonaromatic rather than automatically antiaromatic.

Benzene has six π electrons and is aromatic. Cyclopentadienyl anion has six: four from the two double bonds and two from the anionic lone pair. Tropylium has six from three double bonds while the cationic carbon provides a vacant p orbital. Cyclobutadiene has four π electrons and is antiaromatic. Cyclooctatetraene adopts a nonplanar tub shape and is nonaromatic in its usual ground-state structure.

Count only electrons occupying the cyclic π system. In pyridine, the nitrogen lone pair lies outside that π system; in pyrrole, one nitrogen lone pair supplies two electrons to it. Pyridine can accept a proton on its lone pair while preserving the aromatic sextet. Protonation using the pyrrole lone pair would disrupt the aromatic sextet, helping explain why pyrrole is a much weaker nitrogen base. Do not apply the simple monocyclic counting rule blindly to every fused or three-dimensional aromatic system.

12.3 Tautomerism and keto enol equilibrium

Tautomers are distinct constitutional isomers connected by proton transfer and a change in bond position. They are not resonance contributors because an atomic nucleus moves. A simple keto–enol example is $CH_3 - CHO \rightleftharpoons CH_2 = CH - OH$. Acid or base can catalyse interconversion; the equilibrium composition is determined by the relative free energies of the tautomers in the medium.

In the common keto–enol pathway, an α hydrogen is removed from the carbon adjacent to a carbonyl group. Under base catalysis, the resulting enolate is resonance stabilised: $CH_2^- - CH = O \leftrightarrow CH_2 = CH - O^-$. Protonation at oxygen can give the enol, while protonation at carbon gives the carbonyl compound. Proton removal and proton addition must be shown explicitly; a resonance arrow alone cannot represent tautomerisation.

Many simple aldehydes and ketones favour the keto form. Enol content can be increased by conjugation, favourable intramolecular hydrogen bonding or aromatic stabilisation. Do not assume a fixed percentage for every compound. Ordinary benzaldehyde lacks an α hydrogen for this usual carbonyl α-enolisation pathway; acetone has α hydrogens and can enolise.

12.4 Thermodynamic stability and reaction rate

A more stable product is not necessarily formed faster. Thermodynamic preference concerns relative Gibbs energies and equilibrium, whereas reaction rate depends on the activation free-energy barrier. A stable intermediate may be relatively unreactive once formed, but its formation can still be kinetically accessible or inaccessible depending on the pathway.

A transition state is the maximum along an elementary reaction coordinate and cannot be isolated as a normal molecule. An intermediate is a local energy minimum between elementary steps and may have a finite lifetime. SN2 has one concerted substitution step and no discrete carbocation intermediate. SN1 includes an ionisation step and a carbocation-containing intermediate or ion-pair stage; solvent and substrate strongly influence the rate.

12.5 Substitution elimination addition and rearrangement

Substitution replaces one group with another. Addition consumes a π bond while forming new σ bonds. Elimination commonly removes groups from adjacent atoms to form a π bond. Rearrangement changes the connectivity of the carbon framework or the position of a group within a mechanism. Classify the actual bond changes rather than the reagent name alone.

In E2, base removal of a β proton and leaving-group departure occur in one elementary step. Suitable antiperiplanar geometry is usually important. In E1, ionisation precedes deprotonation and a carbocation can rearrange. Strong bulky bases often favour

elimination over substitution, but the substrate, solvent, leaving group and temperature must be examined together.

A 1,2 hydride or alkyl shift in a carbocation mechanism transfers the migrating group with its bonding electron pair to an adjacent electron-deficient centre. Such rearrangements are often favoured when they lead to a more favourable cation or relieve strain. Do not insert a rearrangement into an ordinary SN2 mechanism, which lacks a free carbocation stage.

12.6 A practical order for solving GOC comparisons

For acidity, identify the exact proton and draw its conjugate base. Compare charge, atom identity, resonance and aromaticity, inductive influence, hybridisation and solvation. For basicity, identify the protonation site and compare the base with its conjugate acid in the stated medium. For nucleophilicity, assess the rate of attack on the specified electrophile, including solvent and steric effects.

For stability of intermediates, identify their charge and electron count first. Draw all relevant conjugation before using the simple tertiary > secondary > primary alkyl-carbocation rule. A resonance-stabilised species is not safely ranked against an unrelated species by one word alone. Always state which structures, medium and mechanism the comparison concerns.

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13 Worked revision questions

Try each question before reading its explanation. These are original revision exercises, not labelled previous year examination questions.

13.1 Acidity and distance

Question Arrange CH_3COOH, $ClCH_2COOH$ and $ClCH_2CH_2COOH$ in decreasing acidity in water.

Answer $ClCH_2COOH$ > $ClCH_2CH_2COOH$ > CH_3COOH. Chlorine stabilises the conjugate base by electron withdrawal, and its influence decreases with distance from the carboxyl group.

13.2 Solvent dependent amine basicity

Question Why can trimethylamine outrank methylamine in the gas phase but lie below it in water?

Answer Gas-phase protonation excludes solvent stabilisation. In water, the relative solvation of each base and its conjugate acid changes the free-energy balance. The standard methylamine series therefore has different orders in the two media.

13.3 Conjugation and hybridisation

Question Why is the nitrogen of an ordinary amide approximately planar?

Answer Its lone pair overlaps with the carbonyl π system. Planarity favours this conjugation, and the $C-N$ bond gains partial double-bond character. A steric-number count that treats the lone pair as wholly localised misses this feature.

13.4 Carbocation stability

Question Compare $CH_3CH_2^+$ with $CH_2=CH-CH_2^+$ qualitatively.

Answer The allyl cation is resonance stabilised because its vacant p orbital overlaps with the adjacent double bond. Ethyl cation lacks this allylic delocalisation. Simply calling both “primary” does not account for the difference.

13.5 Resonance versus tautomerism

Question Are CH_3CHO and $CH_2=CHOH$ resonance contributors?

Answer No. A hydrogen nucleus changes its bonded atom and a π bond changes position. They are keto–enol tautomers, distinct species related by equilibrium.

13.6 Aromatic electron counting

Question Why is cyclopentadienyl anion aromatic?

Answer It can form a planar, continuously conjugated ring with six π electrons: four from two double bonds and two from the anionic lone pair. Six fits 4n + 2 with n = 1.

13.7 Homolysis

Question Identify the products and arrow type for light-induced cleavage of Cl_2.

Answer $Cl-Cl$ gives two $Cl\bullet$ radicals. Two single-barbed fish-hook arrows show one electron moving to each chlorine atom. The reaction arrow alone does not display that electron accounting.

13.8 Heterolysis

Question What products result when both electrons of a $C-Br$ bond go to bromine?

Answer The carbon fragment becomes a cation and bromine becomes Br^- if the starting fragments were neutral. Full-headed electron-pair arrows show the bond pair moving to bromine. Whether free ions form depends on substrate and medium.

13.9 Neopentyl substitution

Question Why is neopentyl bromide slow in an ordinary SN2 reaction despite being primary?

Answer Its adjacent quaternary carbon creates severe crowding for backside approach. Substitution class alone does not capture this β branching. Ordinary ionisation to a primary neopentyl cation is also unfavourable.

13.10 Basicity versus nucleophilicity

Question Why can tert-butoxide be a strong base but a hindered nucleophile?

Answer It readily accepts a small proton, whereas approaching a crowded carbon electrophile is sterically more demanding. With suitable substrates it often promotes elimination.

13.11 Hyperconjugative counting

Question For the elementary α-H counting model, how many adjacent $C-H$ bonds can donate in $(CH_3)_3C^+$?

Answer Nine. The three methyl groups adjacent to the cationic carbon supply nine $C-H$ bonds. This is a counting model for possible donor interactions, not nine separate isolable structures or a complete energy calculation.

13.12 Carbene electron accounting

Question Classify the intermediate produced by $CCl_3^- \rightarrow ? + Cl^-$.

Answer The intermediate is neutral dichlorocarbene, $:CCl_2$. The carbon retains two nonbonding electrons and two $C-Cl$ bonds. The equation conserves charge.

13.13 Carbene stereochemistry

Question What methyl-group relationship is expected when cis-2-butene undergoes concerted singlet-carbene cyclopropanation?

Answer The methyl groups remain cis on the cyclopropane ring. This follows from addition to the same face without rotation about the original alkene bond during the concerted step.

13.14 Grignard work up

Question Complete the net ionic protonation of $\mathrm{RCH_2O^-}$ by $\mathrm{H_3O^+}$.

Answer $\mathrm{RCH_2O^-} + \mathrm{H_3O^+} \rightarrow \mathrm{RCH_2OH} + \mathrm{H_2O}$. Counterions are omitted. This equation conserves atoms and charge.

13.15 Proton transfer equilibrium

Question For HA + $\mathrm{B^-} \rightleftharpoons \mathrm{A^-} + \mathrm{HB}$, estimate K when pK_a(HA) = 5 and $pK_a(\mathrm{HB})$ = 10 in the same solvent.

Answer The relation $\log_{10}K = pK_a(\mathrm{HB}) - pK_a(\mathrm{HA})$ gives $\log_{10}K = 5$, so $K \approx 10^5$. Equilibrium favours the side containing the weaker acid HB and its weaker conjugate base $\mathrm{A^-}$, under the same standard-state convention.

13.16 Halogen directing effect

Question Why is chlorobenzene deactivated yet ortho para directing in electrophilic aromatic substitution?

Answer Chlorine withdraws inductively and reduces overall reactivity, but its lone-pair donation changes the relative stabilisation of the intermediates for attack at different positions. Orientation and overall rate are different comparisons.

13.17 Nitrene spin states

Question How many nonbonding electrons are present on nitrogen in the simple RN nitrene model?

Answer Four. One pair remains paired in the elementary description; the other two electrons distinguish the common singlet and triplet arrangements. Counting only two nonbonding electrons omits a lone pair.

13.18 Mechanistic evidence

Question Does the Simmons–Smith reaction require free methylene?

Answer No. A zinc carbenoid transfers a $\mathrm{CH_2}$ unit to the alkene. Formation of a cyclopropane alone does not identify a free-carbene intermediate.

Further reading and reference data

IUPAC Compendium of Chemical Terminology The Gold Book. Entries on electronic effects and reactive intermediates. https://goldbook.iupac.org/

IUPAC Glossary of Class Names of Organic Compounds and Reactivity Intermediates. https://iupac.qmul.ac.uk/class/ionra.html

NCERT Chemistry Part II Class XI. Organic Chemistry Some Basic Principles and Techniques. https://ncert.nic.in/textbook.php

NIST Computational Chemistry Comparison and Benchmark Database. Experimental geometry records and their original measurement references; distinguish experimental from calculated columns. https://cccbdb.nist.gov/

John McMurry Organic Chemistry OpenStax Section 8.9. Further reading on carbene and carbenoid cyclopropanation. https://openstax.org/books/organic-chemistry/pages/8-9-addition-of-carbenes-to-alkenes-cyclopropane-synthesis

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14 Molecular representation and nomenclature

14.1 Reading a structure

A molecular formula gives the number of atoms of each element; it does not usually specify connectivity. A displayed formula shows bonds explicitly. A condensed formula groups atoms, while a skeletal formula represents carbon atoms at unlabelled line ends and vertices. Carbon-bound hydrogen atoms are normally omitted in a skeletal drawing and are inferred from the normal valence of carbon. Heteroatoms and their attached hydrogens must be shown when required.

For example, $\mathrm{CH_3CH_2OH}$ and $\mathrm{CH_3OCH_3}$ have the same molecular formula $\mathrm{C_2H_6O}$ but different connectivity and functional groups. The placement of parentheses is significant: $\mathrm{(CH_3)_2CHOH}$ means two methyl groups attached to the CH bearing OH. It is not the same structure as $\mathrm{CH_3CH_2CH_2OH}$.

Check the valence of every carbon, oxygen and nitrogen after expanding a condensed formula. A neutral ordinary oxygen commonly makes two bonds; a neutral ordinary amine nitrogen has three bonds and a lone pair. An oxygen with three ordinary single bonds is typically an oxonium centre and must carry a positive formal charge.

14.2 Classifying carbons and functional groups

A primary, secondary, tertiary or quaternary saturated carbon is bonded directly to one, two, three or four other carbon atoms respectively. This classification counts carbon neighbours, not the number of hydrogens in the complete molecule. A primary alcohol has its OH-bearing carbon bonded to one other carbon; methanol is usually treated separately.

Recognise the characteristic groups: alcohol $\mathrm{R-OH}$, phenol $\mathrm{Ar-OH}$, ether $\mathrm{R-O-R'}$, aldehyde $\mathrm{R-CHO}$, ketone $\mathrm{R-CO-R'}$, carboxylic acid $\mathrm{R-COOH}$, ester $\mathrm{R-COOR'}$, amide $\mathrm{R-CONH_2}$, nitrile $\mathrm{R-C\equiv N}$ and amine $\mathrm{R-NH_2}$. The atom through which a group is attached changes its electronic effect: ring–$\mathrm{NHCOCH_3}$ and ring–$\mathrm{CONHCH_3}$ are different substituents.

A homologous series contains compounds with the same characteristic functional group and a common general formula, with successive members differing by $\mathrm{CH_2}$. Their chemical behaviour is related, but boiling points and other physical properties change progressively with size and intermolecular forces.

14.3 A systematic naming procedure

Identify the principal characteristic group, select a parent containing the required group and relevant multiple bonds, and number the parent to give the principal suffix the appropriate lowest locant. Then locate unsaturation and substituents. Choose the parent by the applicable naming rules rather than automatically selecting the visually longest line.

For common acyclic monofunctional examples, assemble substituent prefixes with their locants, the parent stem and the suffix. Alphabetise ordinary substituent names while ignoring multiplicative prefixes such as di and tri for this purpose. Compare locant sequences at their first point of difference; do not use a sum-of-locants shortcut as a universal rule.

A useful selected priority sequence is carboxylic acid > ester > amide > nitrile > aldehyde > ketone > alcohol > amine. This is a selected teaching sequence, not a complete IUPAC seniority table. Halogens and ordinary alkoxy groups are generally prefixes in these elementary examples. A principal alcohol suffix takes numbering priority over an alkene locant in an alkenol.

Examples: $CH_3CH(CH_3)CH_2CH_3$ is 2-methylbutane; $CH_3CH_2CH_2OH$ is propan-1-ol; $CH_3COCH_2CH_3$ is butan-2-one; CH_3CH_2CHO is propanal; $HOCH_2CH_2COOH$ is 3-hydroxypropanoic acid; $CH_2 = CHCH_2OH$ is prop-2-en-1-ol. In CH_3CH_2CN, the nitrile carbon belongs to the parent chain, giving propanenitrile.

14.4 Cyclic and aromatic examples

For a simple substituted cycloalkane, number the ring to obtain the appropriate lowest locant sequence. Identical ring substituents can also generate cis and trans stereoisomers, which are different from merely choosing a numbering direction.

Benzene derivatives are often named with benzene as parent or with a retained parent name such as phenol or benzoic acid. In a disubstituted benzene, 1,2 corresponds to ortho, 1,3 to meta and 1,4 to para. These relative-position terms do not by themselves specify the complete systematic name of a more highly substituted ring.

For example, a methyl group adjacent to the OH-bearing carbon of phenol gives 2-methylphenol; a methyl group opposite it gives 4-methylphenol. In benzoic acid the ring carbon attached to the carboxyl group is position 1.

14.5 Names and structures must agree

A reliable check is to redraw the named compound and then recount every atom and bond. Verify the parent length, principal group, locants, unsaturation and total molecular formula. A change in a locant can change connectivity; a change in a stereochemical descriptor can change spatial arrangement without changing connectivity.

For examination preparation, practise both directions: condensed structure to name, and name to full or skeletal structure. Trivial names such as acetone, acetic acid and toluene should be connected to their structures rather than memorised as isolated words.

15 Isomerism and stereochemical relationships

15.1 Structural isomerism

Structural or constitutional isomers have the same molecular formula but different connectivity. Chain isomerism changes the carbon skeleton, as in pentane and 2-methylbutane. Position isomerism moves a functional group or multiple bond within an otherwise comparable framework, as in propan-1-ol and propan-2-ol. Functional isomerism changes the functional class, as in ethanol and dimethyl ether.

Metamerism is a traditional description of different alkyl-group distributions around a polyvalent functional unit, such as the ether comparison of ethoxyethane with 1-methoxypropane. Ring–chain isomerism is illustrated by propene and cyclopropane, both C_3H_6. These labels describe relationships and should not be counted as entirely independent molecules if the same pair can be discussed in more than one way.

To enumerate isomers, generate distinct carbon skeletons first, place the required functional group at symmetry-inequivalent positions, and eliminate duplicate drawings by renumbering or rotation. Count constitutional isomers before splitting any of them into stereoisomers.

15.2 Geometrical isomerism

Restricted rotation can make different arrangements persistent. For an ordinary alkene to show geometrical isomerism, each double-bonded carbon must have two different substituents. But-2-ene satisfies this condition; propene and 2-methylpropene do not.

Cis and trans descriptors are useful when a suitable pair of comparable substituents can be identified. Cis-but-2-ene has its methyl groups on the same side of the double bond; trans-but-2-ene has them on opposite sides. In rings, restricted motion can similarly distinguish substituents on the same face from those on opposite faces.

Physical-property comparisons require the actual structures. Trans-but-2-ene is usually more stable than cis-but-2-ene because it has less steric crowding, but a universal statement that every trans compound must have a lower boiling point or higher melting point is unsafe. Symmetry, crystal packing and dipole moment affect different properties differently.

15.3 Chirality and one asymmetric centre

A chiral object is not superimposable on its mirror image. A common tetrahedral stereogenic carbon is attached to four different substituents. In 2-chlorobutane, the carbon carrying chlorine is attached to Cl, H, CH_3 and CH_2CH_3, so two mirror-image configurations are possible.

Enantiomers are nonsuperimposable mirror images. In an achiral environment, a pure enantiomer and its partner have the same ordinary melting point and boiling point, but

opposite signs of optical rotation under identical measurement conditions. Their interactions with a chiral reagent, catalyst or biological receptor can differ.

The direction of optical rotation cannot be inferred merely from a drawing of a tetrahedral centre. A racemic mixture contains equal amounts of an enantiomeric pair and has zero net optical rotation because their rotations cancel. A stereogenic carbon is a useful common source of chirality, but chirality is a property of the whole structure, and not every chiral molecule requires such a carbon.

15.4 Two asymmetric centres and meso compounds

For n independent stereogenic centres, 2^n gives an upper bound on the number of configurational stereoisomers before symmetry and other restrictions are considered. It is not a universal final answer. Two unlike stereogenic centres commonly give four stereoisomers, arranged as two enantiomeric pairs.

Diastereomers are stereoisomers that are not mirror images. They generally have different physical properties. A meso compound contains stereogenic centres but is achiral because its entire structure is superimposable on its mirror image. An internal symmetry element can reveal this cancellation, but the comparison must be made using the whole three-dimensional structure.

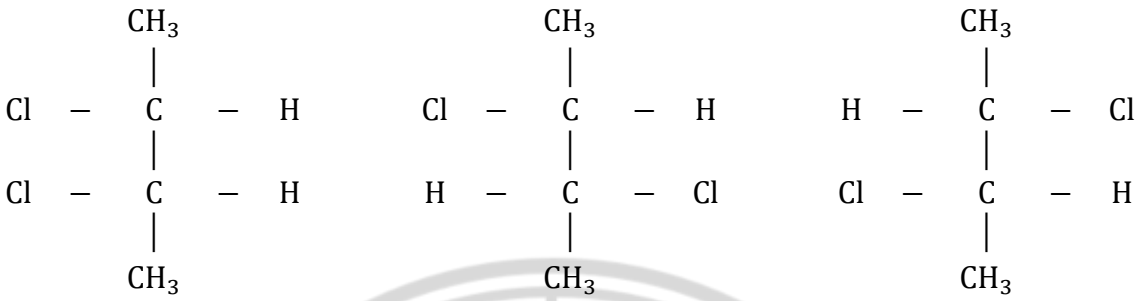
2,3-Dichlorobutane has three configurational stereoisomers: one meso form and an enantiomeric pair. The two ends are identical, so molecular symmetry reduces the four formal assignments to three distinct structures. A meso sample is a single achiral stereoisomer; a racemate is a mixture of two chiral stereoisomers.

15.5 Reading Fischer projections

In a Fischer projection, horizontal bonds point towards the observer and vertical bonds point away. The crossing represents a tetrahedral centre; it is not a planar molecular geometry. Rotating the entire Fischer projection by 180 degrees in its plane preserves configuration. A 90-degree rotation generally does not preserve the Fischer interpretation. Interchanging one pair of groups inverts a single centre, whereas two pair interchanges preserve it.

In the three projections below, identical terminal methyl groups are shown vertically. The arrangement with both chlorine atoms on the same side is meso. The two arrangements with chlorine atoms on opposite sides are nonsuperimposable mirror images of each other. These are configurational comparisons, not different conformations caused by rotation about a single bond.

Meso form Enantiomer A Enantiomer B



15.6 Conformations of ethane and butane

Conformations interconvert by rotation about single bonds without changing connectivity. A Newman projection looks directly along a selected carbon–carbon bond. The front carbon is drawn as a point and the rear carbon as a circle; each contributes three outward bonds. Always identify which atoms belong to the front carbon and which to the rear carbon before measuring a dihedral angle.

For ethane, staggered conformations minimise torsional strain relative to eclipsed conformations. For butane viewed along its central bond, the anti staggered conformation places the methyl groups 180 degrees apart and is the lowest-energy common conformation. Gauche staggered conformations put them 60 degrees apart and are local minima. Eclipsed conformations are higher in energy, with methyl–methyl eclipsing at zero degrees especially unfavourable.

Conformers are not normally counted as separate isolable configurational stereoisomers for these freely rotating small alkanes. Do not confuse anti and gauche conformations with cis and trans configurations of a double bond.

15.7 Syllabus scope for stereochemical labels

This chapter develops structural isomerism, geometrical relationships, enantiomers, diastereomers and meso forms. The official JEE Advanced 2026 basic-principles section specifically excludes R,S and E,Z configuration assignment while including relationships for compounds with up to two asymmetric centres. Treat any additional configuration-assignment training as a separately labelled extension rather than assuming it is required by that particular syllabus statement.

16 Hydrogen bonding and comparison problems

16.1 Intermolecular and intramolecular hydrogen bonding

In common organic examples, an $O-H$ or $N-H$ group can donate a hydrogen bond to an appropriate acceptor lone pair. Ether and carbonyl oxygens can accept hydrogen bonds even though the molecule lacks an $O-H$ bond. An ordinary alcohol can both donate and accept hydrogen bonds. An ordinary amide nitrogen is a poor hydrogen-bond acceptor because its lone pair is involved in carbonyl conjugation.

Intermolecular hydrogen bonding links separate molecules; intramolecular hydrogen bonding occurs within one molecule when the groups can adopt a suitable geometry. Neither description means that the covalent $O-H$ or $N-H$ bond has become a full ionic bond. The strength and extent depend on geometry, solvent and competing interactions.

16.2 Boiling point solubility and association

Among sufficiently comparable small molecules, strong intermolecular association often raises the boiling point. Ethanol can form an extended hydrogen-bonding network, whereas dimethyl ether has no $O-H$ hydrogen-bond donor. Ethers can nevertheless accept hydrogen bonds from water.

Water solubility reflects the balance between polar or hydrogen-bonding interactions and the hydrophobic carbon framework. Adding carbon atoms to a simple alcohol often decreases water solubility despite the continued presence of an OH group. Carboxylic acids may associate as hydrogen-bonded dimers in suitable nonpolar media; the degree of association depends on concentration and solvent.

16.3 Ortho and para nitrophenol

Ortho-nitrophenol can form an intramolecular hydrogen bond between the phenolic group and a neighbouring nitro oxygen. Para-nitrophenol cannot form that same local intramolecular arrangement and can form stronger intermolecular networks. This helps explain their different association and steam-volatility behaviour; ortho-nitrophenol is more readily steam volatile in the standard comparison.

Do not infer all melting points, boiling points, solubilities or acidities from the phrase intramolecular hydrogen bonding alone. Crystal packing, solvent and the relative stabilisation of both the neutral and ionised structures also matter. An acidity explanation must compare the acid with its conjugate base, not merely identify a hydrogen bond in one drawing.

16.4 A hierarchy of checks rather than one ranking trick

For an acidity problem, first draw the conjugate base, then compare the atom carrying charge, resonance and aromaticity, inductive effects and distance, hybridisation, steric

effects and solvation. A lower pK_a in the same medium corresponds to a stronger acid. For basicity, compare the free base and its conjugate acid; for nucleophilicity, compare reaction rates for the specified electrophile and solvent.

Beware of applying a simple alkyl order to allylic or benzylic species, treating every lone pair as equally available, and assuming that steric inhibition of resonance always increases measured basicity. Steric inhibition of solvation can oppose the electronic effect. Draw the specific structures before selecting the dominant explanation.

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17 Formula determination and unsaturation

17.1 Empirical and molecular formulae

An empirical formula is the simplest whole-number atom ratio. A molecular formula gives the actual atom counts in a molecule. The molecular formula is an integer multiple of the empirical formula, with multiplier equal to molar mass divided by empirical-formula mass. Neither formula alone identifies a unique organic structure.

For a compound containing only C, H and O that is completely combusted, all carbon is measured as $\mathrm{CO_2}$ and all hydrogen as $\mathrm{H_2O}$. Using the common examination relative masses 12, 1 and 16, mass of carbon = (12/44) times mass of $\mathrm{CO_2}$ and mass of hydrogen = (2/18) times mass of $\mathrm{H_2O}$. Oxygen in the original sample is obtained by mass difference only when no other elements are present and the data are consistent.

Divide each element mass by its relative atomic mass to obtain relative mole amounts, divide by the smallest, and convert the ratios to small integers. A value near 1.5 requires multiplication of the whole ratio by 2; do not round it directly to either 1 or 2.

17.2 Worked combustion example

Complete combustion of a 0.300 g sample containing only C, H and O gives 0.440 g $\mathrm{CO_2}$ and 0.180 g $\mathrm{H_2O}$. Carbon mass is 0.120 g and hydrogen mass is 0.020 g. The remaining oxygen mass is 0.160 g.

Relative mole amounts are 0.120/12 = 0.010 mol C, 0.020/1 = 0.020 mol H and 0.160/16 = 0.010 mol O. The ratio is 1:2:1, so the empirical formula is $\mathrm{CH_2O}$ and its formula mass is 30. If the measured molar mass is 180 g mol^{-1}, the multiplier is 6 and the molecular formula is $\mathrm{C_6H_{12}O_6}$.

This establishes a formula, not a unique identity such as glucose. Many different structures can have that formula. The calculation assumes complete combustion and a pure sample of the stated elemental composition.

17.3 Degree of unsaturation

For a neutral, conventional closed-shell compound containing carbon, hydrogen, trivalent nitrogen and monovalent halogens, the degree of unsaturation is $\mathrm{DU} = \frac{2\mathrm{C} + 2 + \mathrm{N} - \mathrm{H} - \mathrm{X}}{2}$, where X is the total number of halogen atoms. Oxygen does not enter this elementary count; ordinary divalent sulfur can also be omitted in the conventional formula model.

Each ring or double bond contributes one degree; a triple bond contributes two. Thus $\mathrm{C_6H_6}$ has $\mathrm{DU} = 4$, consistent with three double bonds and one ring in benzene. $\mathrm{C_4H_8}$ has $\mathrm{DU} = 1$ and could be an alkene or a cycloalkane. $\mathrm{C_2H_3N}$ has $\mathrm{DU} = 2$, consistent with the triple bond of acetonitrile. DU narrows possibilities but does not locate the bonds or establish aromaticity.

For ions, radicals, unusual valences or organometallic species, do not apply the neutral closed-shell formula blindly. A negative or fractional value in an ordinary neutral-molecule problem is a warning to recheck the formula, charge or assumptions.

17.4 Worked checks

Question: Determine DU for C_2H_3Cl. Answer: Treat chlorine like hydrogen in the deficiency count. $DU = \frac{2\times2+2-3-1}{2} = 1$; vinyl chloride is one possible structure.

Question: Can C_3H_6 represent a saturated compound? Answer: It can represent cyclopropane, which has only single bonds but one ring. Saturated does not necessarily mean acyclic.

Question: A compound has empirical formula CH_2 and molar mass 56 g mol^{-1}. Find its molecular formula. Answer: The empirical-formula mass is 14, giving multiplier 4 and molecular formula C_4H_8. Its connectivity still requires additional information.

Additional worked questions

Question 1

Name $CH_3CH(OH)CH_2COOH$.

Answer Number from the acid carbon: 3-hydroxybutanoic acid. The acid is the principal suffix and OH is a hydroxy prefix.

Question 2

How many constitutional alcohols have molecular formula $C_4H_{10}O$, ignoring stereoisomers?

Answer Four: butan-1-ol, butan-2-ol, 2-methylpropan-1-ol and 2-methylpropan-2-ol. Butan-2-ol additionally has an enantiomeric pair.

Question 3

Why does but-2-ene show geometrical isomerism but 2-methylpropene does not?

Answer Each double-bonded carbon of but-2-ene has two different substituents. In 2-methylpropene one carbon bears two identical methyl groups and the other bears two hydrogens.

Question 4

How many configurational stereoisomers does 2,3-dichlorobutane have?

Answer Three: an enantiomeric pair and one meso form. Symmetry reduces the maximum count of four.

Question 5

Distinguish a meso compound from a racemic mixture.

Answer A meso compound is one achiral stereoisomer despite having stereogenic centres. A racemate is a 1:1 mixture of two enantiomers; its net rotation cancels externally.

Question 6

Why can dimethyl ether accept a hydrogen bond from water?

Answer Its oxygen has available lone pairs. A hydrogen-bond acceptor need not contain an $\mathrm{O}-\mathrm{H}$ bond of its own.

Question 7

A $\mathrm{C,H,O}$ compound has empirical formula $\mathrm{CH_2O}$ and molar mass 60 g mol⁻¹. Find the molecular formula.

Answer The empirical-formula mass is 30, so the multiplier is two and the molecular formula is $\mathrm{C_2H_4O_2}$. Additional data are needed to identify the structure.

Question 8

What does $\mathrm{DU} = 2$ establish for a neutral ordinary organic molecule?

Answer The total ring-and-pi-bond count is two. Possibilities include a triple bond, two double bonds, two rings, or one ring and one double bond; it does not establish one unique structure.

Syllabus and notation references

Scope reference: JEE Advanced 2026 Basic Principles of Organic Chemistry and the relevant JEE Main 2026 organic-chemistry units. The added chapters cover foundations that complement the earlier electronic-effects chapters; reaction chemistry of individual functional groups requires its own chapter practice.

Official JEE Advanced 2026 syllabus https://jeeadv.ac.in/documents/jee-advanced-2026-syllabus.pdf

Official JEE Main 2026 syllabus https://jeemain.nta.nic.in/document/syllabus-2026/

IUPAC chemical nomenclature and terminology https://iupac.org/what-we-do/books/bluebook/ and https://goldbook.iupac.org/